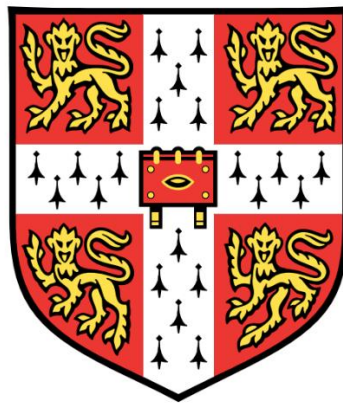


Chomsky's MERGE and the Duality of Semantics

Reconceptualising Copy Visibility under FormCopy



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Declarations

Authorship. This thesis is the result of my own work and includes nothing which is the outcome of work done in collaboration except where specifically indicated in the text.

Statement of length. This thesis is 29999 words in length. Word count includes footnotes but excludes the title page and preface, glosses and translations, figures, and references.

Abstract

This thesis critically examines recent developments in the Minimalist Program, focussing on Chomsky's reconceptualisation of MERGE. I explore the duality of semantics, FormCopy (FC), and their interaction with labelling theory. I discuss two definitions of Duality: EM for theta-roles/argument structure + IM for discourse (Duality₁), and EM for theta-positions + IM for non-theta-positions (Duality₂). I address challenges to both versions, including discourse-related EM, IM unrelated to discourse, and EM to non-theta-positions.

Duality₂ and FC contribute to the copy/repetition/PRO distinction. FC forms copy relations between IM'ed and EM'ed elements, deriving Control. The empirical issues raised by this approach suggest inconsistencies in Chomsky's work concerning which phenomena are visible to the syntax.

The interaction between FC-theory and Preservation, another operation keeping track of copy relations, raises questions. I argue that 'copy' refers to multiple phenomena in MERGE, with inconsistencies about whether copies arise automatically from IM. This relates to questions surrounding the Markovian nature of MERGE, as syntactic operations (including FC) are Markovian, but those pertaining to interpretation (including Preservation) are not.

Lastly, I examine the coherence of these ideas with labelling theory, identifying a conflict between FC-theory and copy invisibility in labelling theory. I propose that labelling conflicts are resolved by FC rather than by movement. Nevertheless, labelling theory presents semantic issues regarding copy invisibility and feature sharing. I adapt Bode's (2020) analysis of copies as a potential solution, but without resolving these issues. I argue that this illustrates a fundamental flaw in labelling theory. Finally, I illustrate the problems arising when unifying FC-theory and labelling by applying my approach to structures which have been analysed in FC-theory.

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Abbreviations

3PL	third person plural
3SG	third person singular
ACC	accusative
AGR	agreement
C-I	the conceptual-intentional interface
DET	determiner
EM	External Merge
EPM	External Pair-Merge
EPP	Extended Projection Principle
EXPL	expletive
FC	FormCopy
FOC	focus
FUT	future
IC	Inclusiveness Condition
IM	Internal Merge
INSTR	instrumental
IPM	Internal Pair-Merge
LA	Labelling Algorithm
LI	lexical item
LSC	language-specific condition
MS	Minimal Search
MY	Minimal Yield
NOM	nominative

NTC	No-Tampering Condition
OC	Obligatory Control
PHC	phase head complement
PIC	Phase Impenetrability Condition
PoP	Problems of Projection
PoP+	Problems of Projection: Extensions
PST	past
Q	question particle
SM	the sensorimotor interface
SO	syntactic object
TOP	topic
vISH	vP-internal subject hypothesis
WS	workspace
WS'	modified workspace

1 Introduction

The quest for theoretical elegance has guided the development of formalisations of minimalist syntax in its search for a more explanatory theory of language.

Accordingly, Chomsky has in recent years proposed a new conceptualisation of Merge, making explicit various elements which so far have been implicit in the theory. This new definition of Merge, sometimes referred to as MERGE, was first introduced in Chomsky's (2019a) Reading lectures, becoming gradually more formalised since (Chomsky, 2019b; 2021; Chomsky et al., 2023). Its immediate predecessor is Problems of Projection (PoP(+)), i.e., labelling theory (Chomsky, 2013, 2015). The transition from labelling theory to MERGE involves a significant theoretical shift, and its implications have not yet been thoroughly explored. This thesis critically examines various key concepts in MERGE, in particular the duality of semantics and FormCopy (FC), and their interaction with labelling theory. This rigorous scrutiny, so far unattempted, is necessary to ensure internal coherence within the theory. This will become evident as I uncover problems and paradoxes inherent in Chomsky's recent work and that of others operating within this framework. I will demonstrate that the theories discussed face several significant issues, with no immediate resolutions in sight. In particular, PoP(+)'s central assumption that lower copies become invisible for labelling is invalid in the MERGE framework, given the introduction of FormCopy.

Before outlining the structure of my thesis, I first provide some theoretical context on Chomsky's MERGE.

1.1 Chomsky's MERGE

MERGE has undergone various developments since its introduction in Chomsky (2019a). This section illustrates MERGE as presented in Chomsky et al. (2023), as this is the clearest, most comprehensive and explicit presentation of MERGE available to date.¹

Chomsky refers to this new version of Merge as MERGE (in uppercase letters) solely in Chomsky (2019b). I use 'MERGE' in this section to highlight the fact that I am discussing a new theory. In the rest of this thesis, however, the terms 'Merge' and 'merged' will be used, with 'MERGE' only used when I focus on the new aspects of Merge; the same operation is being referred to in all these cases.

1.1.1 MERGE applies to a workspace

A key aspect of MERGE is the formalisation of the workspace (WS). The notion of a workspace is mentioned in earlier work (e.g., Chomsky, 2000:106) but does not become an explicit element of the theory until Chomsky (2019a). The workspace contains all elements necessary for computation, serving as the place where all syntactic operations occur. The notion of a workspace is necessary to account for the creation of satellites (e.g., a DP to be merged to a vP). Although being a set, the workspace is not accessible to computation (*pace* Fukuda (2021)).

Lexical items (LIs) are drawn from the lexicon, though the specific details of how this occurs are left open. The workspace contains inscriptions² of these items, as well as

¹ The most recent formulation of MERGE is outlined in Chomsky (2024); however, this version is essentially a slightly modified version of Chomsky (2021).

² Inscriptions refer to the specific instances or tokens of lexical items. For instance, '*One child slept while a second child played with another child.*' (Chomsky et al., 2023:2) contains three inscriptions of the lexical item *child*.

objects created by MERGE. It is updated over the course of the derivation: MERGE takes the WS as input and produces a modified WS' as output:

$$(1) \text{ MERGE (WS)} \rightarrow \text{WS}'^3$$

MERGE targets elements in the workspace, copying them (though see footnote 9) and putting them into a set.⁴ This set is now a member of the WS'. A formal definition of MERGE is presented in (2), with '...' referring to everything in the WS' other than $P_1, \dots, P_m$ (square brackets are used to indicate workspaces, here, the WS').

$$(2) \text{ Merge}(P_1, \dots, P_m, \text{WS}) = \text{WS}' = [\{P_1, \dots, P_m\}, \dots] \text{ (Chomsky et al., 2023:15)}$$

1.1.2 MERGE is binary and subject to Minimal Search

Chomsky et al. (2023) argue that MERGE itself is (2), but is constrained by other conditions, which conspire to produce the desired outcome. This follows the trend of minimising theoretical postulates within UG and accounting for as much as possible using third-factor principles (general principles relating to laws of nature and efficient computation) and interface conditions (the requirements of the conceptual-intentional (C-I) and sensorimotor (SM) interface, i.e., the systems relating to meaning and sound).⁵ Here, I illustrate how third-factor concerns constrain MERGE, making it binary, subject to Minimal Search, and Markovian. A significant interface condition on MERGE, the duality of semantics, is discussed in chapter 2.

³ I use 'workspace' or 'WS' to refer to the pre-MERGE workspace and WS' for the post-MERGE workspace.

⁴ MERGE thus produces copies (e.g., Chomsky, 2019b:44; see chapter 5 for further discussion).

⁵ In Chomsky et al. (2023), interface conditions are referred to as LSCs (language-specific conditions) and are considered first-factor constraints. This is a departure from the standard notion of the first factor consisting only of UG and may be linked to Hauser et al.'s (2002) distinction between FLB (the faculty of language in the broad sense, including interface conditions) and FLN (the faculty of language in the narrow sense, i.e., recursion). Interface conditions, pertaining to FLB, are thus still language-specific conditions. In other words, MERGE is constrained by the first (LSCs) and the third factor. Note also that 'interface requirements' tend to refer to the requirements of the conceptual-intentional interface only, as Chomsky considers this interface primary.

Third-factor considerations of simplicity require MERGE to be binary. Thus, we arrive at (3), which is essentially the standard form of Merge up until Chomsky (2019a), without Binarity needing to be stipulated.

$$(3) \text{ Merge}(P, Q, WS) = WS' = (\{P, Q\}, X_1, \dots, X_n) = \text{Binary Merge}$$

(Chomsky et al., 2023:18)

MERGE is also subject to Minimal Search (MS). In other words, the Search operation must conform to third-factor constraints. It is argued that MS minimises computation. Search operates in steps, first locating an element P from the WS, then locating a Q to merge it with. The first step of Search selects a member⁶ of the workspace. The second step can either search inside P for Q, or select a member of the WS as Q. The first option derives Internal MERGE (IM, i.e., movement), the second produces External MERGE (EM). These are considered suboperations of the same operation Merge since Chomsky (2004b). Up until Chomsky (2013), IM was considered more complex than EM; for instance, in past work, movement was referred to as one of ‘the most striking apparent imperfections of narrow syntax’ (Chomsky, 2000:129). IM applied only once EM was no longer possible. In MERGE, however, IM is considered simpler than EM. Indeed, for EM, the second step of Search requires search of all the members of the WS; whereas IM only requires search into P.⁷ However, since IM and EM are the same operation, there is no notion of EM applying only once IM is blocked (*pace* Rugna & Manzini (2023); see section

⁶ Members of a WS are elements contained directly within it, and ‘X is a *term of* Y iff X is either (i) a member of Y or (ii) a member of a term of Y.’ (Chomsky et al., 2023:19)

⁷ In Chomsky (2019b:23), IM was considered simplest as EM required search of the entire lexicon. This is no longer the case in Chomsky et al. (2023), as there is lexical insertion into the WS, but the general principle still holds.

3.4). Instead, the application of EM versus IM is constrained by Duality, as discussed in chapter 2.

1.1.3 MERGE obeys Minimal Yield

So far, MERGE of a and b is expected to produce (4). Indeed, MERGE targets a and b and combines them into a set, not disrupting the workspace any further.

$$(4) \text{ WS} = [a, b, c]$$

$$\text{MERGE}(a, b, \text{WS})$$

$$\text{WS}' = [\{a, b\}, a, b, c]$$

However, this is undesirable from both a theoretical and empirical standpoint, allowing, for instance, island violations since terms meant to be inaccessible are still available as members of the WS' . Instead, MERGE is restricted by a third-factor principle requiring minimisation of resources. This constraint is based on Fong (2021), who argues that Merge, being a cognitive process, is bound by the brain's resource limitations. In other words, Resource Restriction is a fundamental principle of the human brain, with its linguistic application termed Minimal Yield (MY) (Chomsky, 2019b:42; Chomsky et al., 2023:29).⁸ MY limits the accessible terms that can be produced by MERGE: by MY, MERGE can yield only one accessible term. Thus, merging a and b produces (5), with one new accessible term ($\{a, b\}$), as opposed to (4), which contains three new accessible terms ($\{a, b\}$, a , and b).⁹

⁸ At first only the term Resource Restriction was used (or 'restrict computational resources' in Chomsky (2019a)); MY appeared in Chomsky (2021).

⁹ It is unclear whether MERGE targets a and b , copies them, puts them into a set, and then deletes the original elements; or whether it puts the original a and b into a set. The latter seems simpler, but Chomsky only mentions it in one footnote:

Another, somewhat more intuitive way to think about it is this. Merge targets b and c and puts them into the set $\{b, c\}$ — the inscriptions b, c in $\{b, c\}$ are the original inscriptions from the input WS. Thus, the inscriptions b and c that are members of the output WS (35) are new elements, new inscriptions, and hence new terms. (Chomsky et al., 2023:29).

(5) $WS = [a, b, c]$

$MERGE(a, b, WS)$

$WS' = [\{a, b\}, c]$

MY makes interesting predictions about permissible structures in the grammar.

Indeed, it rules out ‘extensions of Merge’ such as Parallel Merge (Citko, 2005), Sideways Merge (Nunes, 2001) and Late Merge (Lebeaux, 1988). For illustration, Chomsky (2019b:39-40) shows that Parallel Merge of *b* and *c* produces the WS' in (6). Here, two accessible terms are created (the new set and the second occurrence of *b*), violating MY.

(6) $WS = [\{a, b\}, c]$

$MERGE(b, c, WS)$

$WS' = [\{a, b\}, \{b, c\}]$

Internal MERGE does not violate MY, since, although it creates an additional copy of the moved element, MS makes the original copy inaccessible to MERGE.¹⁰ I critically examine this notion in section 5.6.

1.1.4 MERGE is strictly Markovian

Computation is Markovian, which appears to be a general property of computational processes.¹¹ Due to MY, this becomes relevant for language. Syntactic operations, including MERGE, have no memory of past stages of the derivation/ WS . This is

However, these two approaches seem significantly different, so it is strange that they are not discussed in more depth. They have different consequences: MY seems much more relevant to the former than to the latter and only the former fits with MERGE always producing copies (for the latter, this would be the case only with IM). Further research is needed to determine how these models differ in predictions.

¹⁰ Chomsky argues that MY is indifferent to changes in the specific occurrences that are accessible; it focusses solely on the number of accessible terms. The validity of this notion is beyond the scope of this thesis.

¹¹ The Markovian nature of operations is noted in Chomsky (2000).

crucial to Chomsky's approach to copies, repetitions and Control structures, as discussed in chapter 5.

Although MY makes derivations Markovian, Chomsky (2022) introduces the third-factor principle of Preservation, which states that 'the interpretation of an inscription does not change in the course of computation' (Chomsky et al., 2023:22). This is not a syntactic operation and so is not Markovian. I discuss Preservation in chapter 5.

1.1.5 MERGE applies freely

Since PoP+ (Chomsky, 2015) Merge (both IM and EM) applies freely, with ungrammatical structures filtered out by the interfaces or barred by third-factor constraints. However, the discussion around this is somewhat confusing. Indeed, the third-factor elements of Binarity, MY, and MS are considered constraints on the application of MERGE, hence the difference between (2) and (4), implying that MERGE is not entirely free (in the sense of 'unconstrained'). Furthermore, the status of interface requirements is less clear. Section 3.6 discusses the ambiguity of whether Duality should be considered a constraint or a filter, and section 6.2 questions whether labelling is a trigger or a filter. Indeed, a recurring theme in this thesis is the implication for the interfaces of attempting to simplify MERGE as much as possible by relegating many elements to the semantics. This leads to inconsistencies about whether certain elements belong to the interfaces or operate within syntax.

1.1.6 Other operations

This section briefly discusses the status of Pair-Merge and head movement in MERGE. Detailed discussions of these phenomena are beyond the scope of this work, but a brief presentation is necessary for context.

As seen in section 1.1.3, MY rules out extensions of Merge. However, one additional operation is present in the theory, namely Pair-Merge, which Chomsky argues is necessary to account for adjuncts. Many questions follow, including how the system knows when to use Pair-Merge and the nature of the dimension in which Pair-Merge applies. Section 2.5 discusses Pair-Merge in the context of Duality. Ideally, Pair-Merge would be eliminated from the theory as it is a significant complication to MERGE. For approaches to adjuncts in minimalism, see Bode (2020).

Since Chomsky (2001), Chomsky has regarded head movement as a part of externalisation.¹² Recently, Chomsky has argued that head movement is countercyclic (Chomsky et al., 2019:249; Chomsky, 2021:30), presumably because it involves forming an amalgam with a previously merged element. However, disallowing IM of heads seems stipulative. For discussion of head movement in minimalism, see Bode (2020:137) and references therein. Section 6.6 mentions some labelling issues faced by head movement.

1.2 Thesis roadmap

The previous section outlines the MERGE framework, which has several implications. This thesis focusses on two main areas: a critical discussion of the duality of semantics, and the implications of the FormCopy operation for copies and labelling. Duality and FormCopy are interlinked in Chomsky's work, as these are both used to derive copies and Control structures.

Chapters 2-4 discuss the duality of semantics, a concept which has existed since before Chomsky (2019b), but which has gained particular significance in MERGE.

¹² However, it is unclear how this aligns with the fact that he sometimes derives head movement via Pair-Merge (e.g., in PoP(+)).

Duality has been defined in various ways. Chapter 2 considers the earlier definition, which I refer to as Duality₁: EM for theta-roles/argument structure + IM for discourse. I discuss challenges to this, including discourse-related EM and IM unrelated to discourse. This version of Duality has also been linked to the A/A' distinction, raising numerous questions which I also address.

Chapter 3 explores Duality₂, the later version of Duality: EM for theta-positions + IM for non-theta-positions. This is derived from Preservation and theta-theory and is argued to constrain Merge. Challenges to this approach include EM of non-arguments, and the fact that theta-roles, being assigned/interpreted configurationally, cannot constrain Merge. Chomsky also attempts to link Duality₂ to the A/A' distinction, though in a different way to Duality₁, which presents its own challenges.

Duality₂ and FormCopy contribute to the copy/repetition/PRO distinction: this is the focus of chapter 4. After briefly discussing past approaches to the copy/repetition distinction, I examine how this is done in MERGE. This involves the introduction of FC: FC can form copy relations between IM'ed and EM'ed elements, which, alongside Duality₂, derives Control. I discuss various empirical issues with this definition of Control, arguing that Chomsky is inconsistent about what is considered visible to the syntax.

After presenting empirical issues with FC in chapter 4, chapter 5 turns to theoretical issues surrounding FC, including its interactions with Preservation, another operation relevant to copies. I argue that Chomsky is inconsistent about whether IM-copy relations come for free in MERGE. Indeed, with FC, it is claimed that copies no longer arise automatically, yet IM-copy relations remain through Preservation. This leads to inconsistencies concerning copy definition, with a huge proliferation of

elements referred to as ‘copies’ appearing in the theory. This is further complicated by the various interpretations others have assigned to Chomsky’s work, given its inherent ambiguity. FC-theory also presents issues concerning accessibility of elements to MS. Lastly, FC and Preservation relate to questions surrounding the Markovian nature of MERGE and the significant difference that now arises between representational and derivational perspectives on the same structure in Markovian MERGE.

Chapter 6 discusses the implications of FC on labelling theory. I identify a conflict between FC-theory and lower copy invisibility in PoP(+), and propose that labelling conflicts are resolved by FC rather than by movement. I discuss other issues facing labelling theory, focussing on the overarching theme that labels are now viewed as semantic, despite primarily being useful to the syntax. I attempt to adapt Bode (2020) and explore additional approaches to address this issue, but conclude that this illustrates a fundamental flaw of labelling theory. The final section illustrates the problems which arise when attempting to unify labelling theory and FC-theory. I present derivations of structures which have been analysed using FC, adapted to my approach. My model makes similar empirical predictions to PoP(+), but the specific examples in question and labelling theory in general both face significant issues. Thus, although labelling theory and FC-theory can be unified somewhat successfully, the fundamental flaws with both theories remain, with no ready solutions forthcoming.

2 The duality of semantics (Duality₁)

Broadly summarised, the duality of semantics states that External Merge expresses argument structure and Internal Merge the other properties, in particular discourse-related ones. Duality was briefly mentioned in Chomsky's earlier work (e.g., Chomsky, 2000; 2004b; 2008), but its relevance to the theory has become more apparent in Chomsky's latest formulation of Merge (Chomsky, 2019b; 2021; Chomsky et al., 2023). In this chapter, I examine the evolution of Chomsky's perspective on Duality until Chomsky (2019b) and critically evaluate its empirical predictions with a focus on discourse.

2.1 Origins of the duality of semantics

In the early stages of generative grammar (e.g., Chomsky, 1965), deep structure determined semantic interpretation, whereas surface structure determined phonetic realisation. Later (Chomsky & Lasnik, 1993), D-structure fed S-structure, which was the output of syntax, sent to both PF and LF¹³. In minimalism, the D/S-structure divide is eliminated and there are no longer levels in the syntax (Chomsky, 1995b:ch3).

Nevertheless, Chomsky's early minimalist work (e.g., Chomsky, 2000:120-121,127) refers to ideas about the distinction in meaning between the two, namely that D- and S-structure correlated with different types of semantic interpretation.¹⁴

One approach [...] was to distinguish the role of deep and surface structure (D- and S-structure) in semantic interpretation: the former enters into

¹³ i.e., Phonological Form and Logical Form, levels of representation associated with sound and meaning respectively, now obsolete in minimalism.

¹⁴ These early ideas preceding the duality of semantics originated in the work of Jackendoff (1969), Culicover (1970), Jenkins (1972) and others.

determining quasi-logical properties such as entailment and θ -structure; the latter such properties such as topic-comment, presupposition, focus, specificity, new/old information, agentive force, and others that are often considered more discourse-oriented and appear to involve the "edge" of constructions. (Chomsky, 2000:120-121)

Chomsky revisits these concepts due to concerns about movement. As mentioned in section 1.1.2, movement was considered a significant complication to the theory, requiring justification. External systems of thought require both types of meaning to be generated; Move is necessary to express some elements of semantic interpretation distinct from Merge, namely those related to discourse. Although we cannot equate D-structure with Merge and S-structure with Move, as the system is now fundamentally distinct, the overall goal is to achieve this semantic mapping, and thus to justify the existence of dislocation.

In Chomsky (2000), this division of roles between Merge and Move is mostly a conjecture about the thought system. If the thought system functions this way, Move and Merge are optimal for expressing these meanings (with Chomsky admitting that the precise division of semantic properties is not quite clear (Chomsky, 2002:113-114)).

2.2 Duality₁: IM for discourse, EM for argument structure

The term 'duality of semantics' was first used in Chomsky (2004b:165), with Chomsky (2004a:110) referring to the 'duality of semantic interpretation at SEM'. Here, Chomsky argues that the multiplicity of semantic meanings referred to in section 2.1 can be reduced to a duality, namely argument structure and everything else (edge properties, in particular scope and discourse; examples to be discussed

in section 2.3). This clarifies previous work, specifying that the thought systems (at this point, SEM) require this duality: EM and IM¹⁵ are constrained by an interface requirement. For instance, in Chomsky (2004a:112-113), the EPP-feature (here ‘OCC’) which triggers movement should be available only when necessary, i.e., when movement is required to express a meaning not expressible by EM: OCC is only available when it yields scope- or discourse-related properties.

Notably, the meanings associated with IM are unclear. IM is said to yield both discourse-related properties and everything except argument structure. In Chomsky (2008), IM is said to yield ‘discourse-related properties such as old information and specificity, along with scopal effects’ (p.140), but also ‘semantic properties apart from generalized argument structure’ (p.141) more generally.

2.3 Empirical challenges to Duality₁

So far, Duality₁ has been defined as follows:

- (7) EM yields argument structure, IM yields everything else.
- (8) EM yields argument structure, IM yields discourse/scope/edge properties.

In this thesis, ‘discourse’ is defined as relating to phenomena operating beyond the sentence level. I focus on information structure, i.e., how language encodes the status of information in discourse, with an emphasis on topic and focus.

Chomsky does not provide examples when discussing Duality₁. This chapter examines possible challenges to Duality₁, namely cases of EM associated with discourse. I focus on discourse particles, arguing that the discourse-syntax mapping seems more complex than (7)-(8) suggest.

¹⁵ Since Chomsky (2004b), Move and Merge are replaced by EM and IM.

Other empirical challenges to Duality₁ are discussed in later sections: the base-generation of elements in the left periphery (i.e., a discourse position), as in left dislocation structures, is explored in chapter 3, as they are particularly relevant to Duality₂. IM unrelated to discourse, a challenge for (8), is discussed in section 2.4.

Chomsky argues that while Duality is not a logical necessity, it is the optimal outcome: IM and EM come freely, so having them express this interface requirement is optimal. However, there is nothing stopping a language from utilising other devices to express Duality, for instance externally merging a particle to express discourse meaning (Chomsky, 2004a:111; 2004b:165). This common phenomenon occurs in Japanese, for instance: examples (9) and (10) carry different discourse meanings due to the different particles involved, yet the word order and propositional content remain the same.

- (9) Jonsan-ha neko-wo katteimasu
 John-TOP cat-ACC has
 ‘John has a cat’

- (10) Jonsan-ga neko-wo katteimasu
 John-NOM cat-ACC has
 ‘It’s John who has a cat’

There are also instances where movement and discourse particles co-occur, as in (11).

- (11) neko-ha Jonsan-ga katteimasu
 cat-TOP John-NOM has
 ‘As for the cat, John takes care of it’

Chomsky (2004b:165) argues that discourse particles are secondary for discourse meaning. Apparent exceptions to Duality (such as the distinction between (9) and (10)) are argued to involve covert movement. For instance, Chinese *wh*-interrogatives are standardly analysed as involving covert movement (Cheng, 1991): for Chomsky, the primary source of discourse meaning in (12) is covert movement, rather than the question particle *ne*.

(12)	ni	xihuan	shenme	dongwu	ne
	you	like	which	animal(s)	Q

‘Which animal(s) do you like?’

Examples such as (11)-(12) demonstrate the existence of structures involving both (overt or covert) movement and discourse particles. However, whether the discourse particle is merely a reflex of movement or has a more significant role is up for debate. Indeed, Aboh (2010) argues the opposite of Chomsky, claiming that discourse features (e.g., Interrogative Force, Topic, Focus), which are added in the numeration¹⁶, are the source of discourse meaning, with movement being the reflex. Thus, sentences like (13) might involve a covert discourse particle.

(13) Which animals do you like?

Aboh argues that information structure must begin in the numeration. He shows that discourse information is relevant to the syntax in various languages; however, adding

¹⁶ Chomsky (1995b:207) introduced the notion of a numeration, i.e., a set containing the LIs, selected from the lexicon, which are used in syntactic derivations. Numerations also include indices specifying how many times each LI has been selected. Chomsky stopped mentioning numerations after Chomsky (2004a).

this information later would violate the Inclusiveness Condition.^{17 18} Discourse features are present on discourse particles, and are also added to LIs, in the same way as Case and phi-features for example are.¹⁹

Aboh relies heavily on Gungbe, an SVO Niger-Congo language of the Kwa family, which contains both discourse-related and T-related particles. Aboh argues that discourse features behave like other syntactic features. For instance, (14)-(16) illustrate that Gungbe T-related particles behave like discourse-related ones. Subject Raising is mandatory with both the future marker *ná* and the topic and focus particles *yà* and *wè*. Although there is movement here, it is not the cause of the discourse meaning but rather a reflex of the particle.

- (14) a. Kòfí ná xò xwé ló
 Kofi FUT buy house DET
 ‘Kofi will buy the house’
- b. *ná Kòfí xò xwé ló
 FUT Kofi buy house DET
- c. *xwé ló ná Kòfí xò
 house DET FUT Kofi buy

¹⁷ The Inclusiveness Condition (IC) states that no new objects can be added during computation. Structures consist only of the elements present in the input, namely LIs. This bars for instance traces or indices (Chomsky, 1995b:209).

¹⁸ Thus, this presumably contradicts configurational approaches to theta-role assignment such as that of Chomsky (see section 3.6). Indeed, configurational theta-role assignment would presumably also violate the Inclusiveness Condition. Theta-roles would instead have to be features assigned in the numeration.

¹⁹ Probe-Goal interactions, analogous to Agree, take place between the particles and the LIs, creating the relevant discourse meanings.

(15) a. Kòfí wè xò xwé ló
 Kofi FOC buy house DET
 ‘KOFI bought the house’

b. *wè Kòfí xò xwé ló
 FOC Kofi buy house DET

(16) a. Kòfí yà é xò xwé ló
 Kofi TOP 3SG buy house DET
 ‘As for Kofi, he bought the house’

b. *yà Kòfí xò xwé ló
 TOP Kofi buy house DET

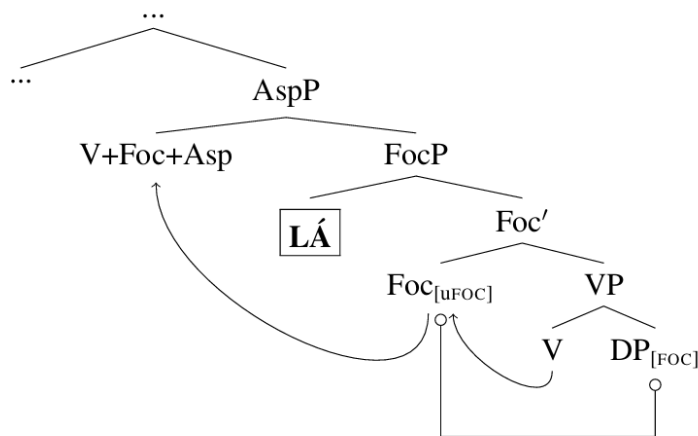
(Aboh, 2010:36-37)

Thus, even in structures with discourse-related movement, this may not necessarily be the sole or primary source of discourse meaning; however, a detailed critique of Aboh’s approach and its implications is beyond the scope of this work.

Lastly, I provide a final example of a case where discourse meaning is seemingly solely conveyed by a particle. Mursell et al. (2022) argue that the Mabia language Dagbani has an in-situ focus marker which is EM’ed in the vP periphery (i.e., in the low focus projection as introduced by Belletti (2004), a position paralleling Rizzi’s (1997) focus position in the left periphery).

The Dagbani in-situ focus marker /á appears in a fixed position in the clause, following the verb. Mursell et al. analyse in-situ focus as (17), with circles indicating agreement between the FOC head and the focalised DP.

(17)



(Mursell et al., 2022:7)

Here, it appears that discourse meaning is realised by the particle rather than by movement. However, nothing inherently prevents an analysis involving covert movement to the low focus projection to appeal to Duality₁, though such an approach may lack economy. It remains unclear what kind of evidence could definitively rule out the possibility of covert movement, rendering this notion somewhat unfalsifiable. Even if covert movement is posited, however, it is dubious that this movement would serve as the primary source of discourse meaning rather than being a reflex of the particle.

In conclusion, discourse meanings seem to be realised by either EM'ed particles, movement, or an interaction of the two. Analysing discourse meaning as involving underlyingly only movement (or only particles) is problematic: in some cases, discourse meanings seem to be realised by particles (e.g., Dagbani), with any (overt or covert) movement being a reflex (e.g., Gungbe, according to Aboh). In other cases (e.g., English), movement seems to be the source of discourse, with any (overt or covert) particles being the reflex (e.g., Chinese, according to Chomsky). Thus, although the relative importance of particles and movement in contributing to

discourse meaning remains debated, it appears that they can both convey discourse meaning, suggesting that Duality₁ as in (7) and (8) is too strict. Even the existence of discourse particles EM'ed into the structure (whether or not they are the primary conveyor of this meaning²⁰) is problematic, if we claim that EM is always related to argument structure; although Chomsky (2005:14) concedes that the role of EM can extend to include functional categories such as those of the cartographic tradition (e.g., Cinque, 2002; Rizzi, 2004; Belletti, 2004). More broadly, however, clearly not everything that is EM'ed is related to argument structure, as not all EM'ed elements are arguments. I return to this in section 3.3.2.

2.4 Duality₁ and the A/A' distinction

Chomsky (2007, 2008) posits that Duality is associated with the A/A' distinction, i.e., the fundamental difference, formulated in Chomsky (1981), between A-positions and A'-positions in the syntax. A-positions include theta-positions as well as those associated with Case. A'-positions are associated with (wh- and quantifier-related) scope and discourse. A- and A'-positions are argued to behave differently and form distinct domains that resist interactions, e.g., in terms of movement and intervention effects.

Chomsky (2008) argues that the A/A' distinction, which he derives from feature inheritance (a now obsolete notion introduced in the same paper, whereby features on phase heads can be passed down to their complements), is necessary due to a C-I requirement for both arguments and quantification, which he suggests are expressed by operator-variable structures. This requirement is 'analogous to the requirement of semantic duality that is satisfied in an optimal way by the A-A'

²⁰ Indeed, if they contribute to discourse meaning, they violate Duality₁, regardless of whether they are the primary source of such meaning.

distinction' (Chomsky, 2008:148), with operator-variable constructions²¹ 'virtually [coming] free, given EM and IM expressing the duality of semantics at CI' (Chomsky, 2007:12). He continues:

'In the simplest case, the copy merged to the edge by IM is the operator taking scope over the copy that had previously been merged by EM, the latter understood as the variable; the full structure of the two copies provides the interpretation as a restricted variable [...] These considerations take us a step towards establishing the A/A' distinction as a property of language with a principled explanation in terms of SMT.' (Chomsky, 2007:12)

2.4.2 A-movement

However, the link between the A/A' distinction and Duality is not obvious. The idea that A'-movement yields operators which are associated with scope, discourse etc. is consistent with IM yielding discourse-related meanings but does not account for A-movement. Chomsky seems to associate the A-system solely with EM:

'The product of A-movement is an expression with one or more copies of an element *that initially entered the computation by EM, receiving its role in argument structure*. The collection of these copies is an A-chain [...] One way to yield such interpretations [of quantification] "comes free," without stipulation: an operator in SPEC C taking scope over *an A-chain regarded as a variable*' (Chomsky, 2007:24, emphasis mine)

Here, focus is given to the lowest copy of an A-chain being EM'ed, thus associated with argument structure, aligning with Duality₁. However, the entire A-chain is

²¹ The status of operator-variable constructions in the theory is problematic, not least because copies are supposed to be identical (for more discussion on copy identity, see chapter 5). A criticism of operators is beyond the scope of this thesis; for discussion, see Fox (1999).

considered the variable, i.e., the element associated with argument structure, even though movement may have been involved. In disregarding A-movement, Chomsky ignores the T-domain. Standard clausal structure is comprised of v-, T- and C-domains; however, in his work on Duality, when referring to the A-system, Chomsky focusses solely on theta-positions (the v-domain) and ignores Case-positions (the T-domain).

This version of Duality₁ is summarised in (18):

- (18) EM (and IM?) in the A-system yields argument structure, IM in the A'-system yields discourse.

A-movement is typically considered unrelated to discourse, for instance subject movement to Spec-TP. However, where it is optional, as in Icelandic, e.g., (19), or Russian, e.g., (20), it can take on discourse significance.

- (19) a. Það hefur verið einhver köttur í eldhúsinu
 EXPL has been some cat in kitchen.the
 b. Það hefur einhver köttur verið í eldhúsinu
 EXPL has some cat been in kitchen.the
 Both: 'There has been some cat in the kitchen' (Svenonius, 2002:44)

- (20) a. Soldata ranilo pulej
 soldier-ACC wounded_[-AGR] bullet.INSTR
 b. Pulek ranilo soldata
 bullet.INSTR wounded_[-AGR] soldier-ACC
 Both: 'A soldier was wounded by a bullet' (Lavine & Freidin, 2002:280)

Although passives are classically analysed in generative syntax as A-movement unrelated to discourse (e.g., Baker et al., 1989 et seq.), they also involve a discourse

component. Further discussion is beyond the scope of this work, but if discourse-related A-movement is indeed possible, A-movement is both evidence and counterevidence for Duality₁, depending on the version considered. Discourse-related A-movement is consistent with (8). Such movement is a problem for (18), however, since the A-system (both EM and IM) is supposed to be associated with argument structure. Conversely, A-movement independent of discourse is unproblematic for (18) but provides counterevidence for (8).²²

This version of Duality₁, whereby Chomsky tries to maximise theoretical elegance by aligning the A/A' distinction, operator-variable constructions, and Duality, is questionable. The differing predictions between (8) and (18) highlight the uncertain nature of A-movement in a framework where the T-domain is ignored.

2.4.3 Duality₁ and phases

Chomsky et al. (2019) also attempt to unify Duality₁ with the A/A' distinction by stating that EM within the vP-phase provides argument structure and IM at the CP-phase expresses scope- or discourse-related properties (Chomsky et al., 2019:241-242). This is reminiscent of Chomsky (2007, 2008), since A'-movement happens essentially at the CP-phase and the initial theta-component of the A-system is associated with the vP-phase.²³ This definition is more restrictive with its reference to specific phases, but has no empirical or theoretical justification; the theoretical

²² Other potential counterevidence for (8), i.e., movement unrelated to discourse, includes head movement and movement approaches to crosslinguistic word order variation as in Kayne (1994). For Chomsky, both phenomena are relegated to externalisation (see section 1.1.6 on head movement; parameters are considered a part of externalisation (e.g., Chomsky, 2021:11-12)).

²³ It is also reminiscent of Chomsky (2007) where he states that 'If the conclusions about duality of semantics and the IM-EM distinction are correct, then C must be the locus of A'-movement to capture scopal properties, so that the phasal character of CP may follow directly from CI-interface conditions and optimal computation, hence SMT' (Chomsky, 2007:17), where he appears to be attempting to derive phasehood from Duality₁.

elegance of aligning Duality with the two (clausal) phases is not a convincing argument in isolation.

In particular, the status of IM at the vP-phase and EM at the CP-phase is left unclear. EM at the CP-phase includes discourse particles merged in the left periphery, such as topic and focus heads in Rizzi's (1997) approach (e.g., as in (11)), as well as left dislocation, which I discuss in section 3.4.1.

IM at the vP edge is particularly problematic, as the vP edge is the escape hatch for elements wishing to exit the vP-phase. For instance, wh-movement must move through the vP edge to adhere to the Phase Impenetrability Condition (PIC).²⁴ Furthermore, the vP edge contains a low focus position in cartographic approaches (Belletti, 2004). In such approaches, a strict division whereby vP yields argument structure and CP discourse is impossible. Chomsky's later work lacks this conflation of Duality with the CP- and vP-phases.

2.4.4 The A/A' distinction and contextuality

Another concern with the versions of Duality₁ discussed here is whether the A/A' distinction is real. Chomsky's views on this shift regularly; Chomsky et al. (2023:63) argue the distinction is eliminable.

Although critical discussion of the A/A' distinction is beyond the scope of this work, it is worth noting Bošković's work on contextuality (see also section 5.7.1). Bošković (2024) argues that many aspects of syntax are contextual, including the notion of phases and the A/A' distinction. For instance, he argues that the landing site for

²⁴ The PIC is a locality restriction stating that certain elements are inaccessible due to having been transferred to the interfaces at the phase level. Originally posited in Chomsky (2000:108), it has been defined in multiple ways. In this thesis, I use the version used in MERGE, which takes the phase head complement to be inaccessible to operations outside HP, while the head H and edge remain visible (e.g., Chomsky et al., 2023:38). This is sometimes referred to as PIC₁ (e.g., Citko, 2014:33).

subject wh-movement is a position somewhere between CP and TP, which cannot be neatly classified as either A or A', and that this is also the landing site for focus-movement. If this approach is correct, associating Duality with the A/A' distinction or specific phases (since these are contextual too) is suboptimal.

I return to the A/A' distinction in section 3.4 in the context of Duality₂.

2.5 Pair-Merge

Having discussed various versions of Duality₁ and their empirical challenges, one question remains: the nature of the relationship between Pair-Merge and Duality. As discussed in section 1.1.6, Pair-Merge remains incorporated in Chomsky's work. If IM yields discourse and EM argument structure, what semantic information is provided by Pair-Merge?

Chomsky (2004a) mentions Pair-Merge in the context of Duality: it is considered necessary to yield adjuncts, expressing predicate composition in the semantics. (Set-)Merge, which yields Duality, cannot do this.

Possibly richness of expressive power requires an operation of predicate composition: that is not provided by set Merge, which yields the duality of interpretation discussed earlier: argument structure and edge properties. But it is the essential semantic contribution of pair Merge. (Chomsky, 2004a:118)

Richards (2009) argues that Pair-Merge can be separated into Internal and External Pair-Merge, analogously to Internal and External (Set-)Merge. The implications of this division on Duality are unclear. One possibility is to suggest that EPM yields predicate composition, whereas IPM serves a different function. Richards argues that IPM yields discourse-driven stylistic movement like topicalisation, focus-movement and scrambling. It is unclear how these components would align, and

beyond the scope of this thesis to determine whether discourse-driven movement should be analysed as Pair-Merge, which seems contradictory to Duality₁. However, such notions are worth exploring if we aim to retain Pair-Merge within the theoretical framework. For relevant discussion of ongoing debates concerning adjuncts, see Bode (2020).

2.6 Summary

The duality of semantics has been defined in various ways. It evolved from a conjecture about the thought system used to justify the existence of Move, to an interface requirement constraining IM and EM. It has been associated with the A/A' distinction and even with phases.

The main variations of Duality₁ are summarised here:

- (21) EM yields argument structure, IM yields everything else. (= (7))
- (22) EM yields argument structure, IM yields discourse/scope/edge properties.
(= (8))
- (23) EM (and IM?) in the A-system yields argument structure, IM in the A'-system yields discourse. (= (18))
- (24) EM within the vP-phase provides argument structure, and IM at the CP-phase expresses scope- or discourse-related properties.
- (25) EM yields argument structure, IM yields discourse; EPM yields predicate composition, IPM yields discourse.

Section 2.3 discussed empirical challenges to (7) and (8) in the form of discourse particles, arguing that Duality₁ is too strict, as discourse is realised by either particles, movement, or an interaction of the two. Regardless of whether movement is involved in all cases of discourse meaning, the existence of EM'ed discourse particles poses a problem to Duality₁.

Section 2.4 discussed the relevance of the A/A' distinction to Duality₁, with which, in (23), Chomsky aims to conflate Duality₁ to gain theoretical elegance. Since Duality is considered an interface condition, this is perhaps an attempt at explanatory adequacy; however, empirical and theoretical issues abound. I argue that Chomsky overlooks the T-domain, resulting in a failure to account for A-movement, which can be discourse-related but does not have to be. This has different implications for (22) and (23).

(24) aims to conflate Duality with phases. Issues include the fact that vP provides an escape hatch for movement and the existence of the low focus projection in vP. Furthermore, the status of the A/A' distinction remains unclear in Chomsky's own work, and contextual approaches to syntax suggest it may not be real. Lastly, the implications of Pair-Merge for Duality₁ remain unexamined.

Until this point, Duality has primarily been posited to justify the existence of movement. Duality₁ also constrains the application of Merge, but this has not yet been exploited seriously in the theory. Duality remains a vague notion which has not been formalised. In the following chapter, I discuss the duality of semantics in Chomsky's new approach to Merge, where it is made more specific and given a much bigger and more concrete role.

3 The duality of semantics in MERGE (Duality₂)

Chomsky is silent on Duality during the labelling era. It reappears in Chomsky (2019b) and takes on new importance in the MERGE framework, where it becomes relevant for the copy/repetition distinction, the trace/PRO distinction, and generally to systematically constrain the application of Merge. This section examines the various definitions and uses of the duality of semantics in MERGE, which I refer to as Duality₂.

3.1 The UCLA Lectures

In the UCLA Lectures (Chomsky, 2019b), Duality is made more specific. If EM generates argument structure (e.g., theta-roles) and IM generates discourse-related properties, then theta-positions can be filled only by EM.²⁵ Duality is thus defined as (26):

(26) EM to theta-positions, IM to non-theta-positions.

In Chomsky (2019b), Duality is still part of how Chomsky justifies the existence of both types of Merge; however, this time EM requires justification, since IM is now considered simpler than EM (see section 1.1.2).

Duality₂ is used to distinguish copies and repetitions, which I discuss in chapter 4.

3.2 Deriving Duality₂

In Chomsky et al. (2023), Duality₂ as in (26) is derived from the interaction between Preservation and theta-theory. Preservation states that ‘the interpretation of an

²⁵ This was alluded to in Chomsky’s (2000) conception of the theta-theoretic principle, where he stated that theta-positions can only be filled through (External) Merge of an argument: ‘Pure Merge in θ -position is required of (and restricted to) arguments.’ (Chomsky, 2000:103). This builds on ideas from the D-structure era, as discussed in section 2.1. Indeed, as we shall see, Duality and theta-theory are closely linked in later work.

inscription does not change in the course of computation' (Chomsky et al., 2023:22).

Theta-theory²⁶ concerns thematic relations between arguments and predicates. The Theta-Criterion from Chomsky (1981) is carried over into MERGE, as in (27):

- (27) every theta role must be assigned to one, but only one argument; and every argument must receive one, but only one theta role. (Chomsky et al., 2023:31)

Because of Preservation, IM to theta-positions is disallowed. Movement of an element to a theta-position would alter its interpretation²⁷, as it gains a theta-role. IM is thus restricted to non-theta-positions.

Due to theta-theory, EM to non-theta-positions is disallowed. Arguments must receive a theta-role and cannot move to receive it as this would violate Preservation, as noted above. Thus, they must receive a theta-role in their EM'ed position. EM of arguments is therefore restricted to theta-positions.

Together, these constraints derive Duality₂: the only permitted cases of EM are those to theta-positions, and IM is restricted to non-theta-positions.

This differs from previous formulations of Duality which did not try to explain or derive it, but purely stated it as an interface condition. Deriving Duality from simpler principles seems desirable.²⁸ This accounts for criticisms such as Ortega-Santos (2012:237) which suggest it is stipulative. Duality is still referred to as an interface condition and its status as its own operation is sometimes unclear, but this appears to be mainly a terminological confusion.

²⁶ Theta-theory was associated with Duality in Chomsky (2021), but the link between the two is made clearer here.

²⁷ i.e., theta-interpretation. The specific meaning of 'interpretation' here is discussed in section 5.3.2.

²⁸ It seems to me that positing Preservation is less stipulative than positing Duality, so deriving Duality from Preservation might indeed be a simplification.

Duality now plays a significant role, reducing options in the application of Merge of arguments, and segregating IM and EM, which Chomsky et al. (2023:33) argue increases simplicity and economy.

3.3 Empirical challenges

(26) makes specific predictions about when EM and IM may apply. This section discusses possible challenges to Duality₂, namely IM to a theta-position and EM of non-arguments.

3.3.1 IM to a theta-position

The most cited example of IM to a theta-position is Hornstein's (1999) movement theory of Control, which essentially argues for the same derivation for Raising and Control structures. In other words, (28) and (29) are both derived via movement. Crucially, (28) involves movement to a theta-position.

(28) The cat wants ~~the cat~~ to eat a fish – Control

{_{VP} **the cat** {_{VP} wants {the cat to eat a fish}}}

(29) The cat seems ~~the cat~~ to be happy – Raising

{_{VP} **the cat** {_{VP} seems {the cat to be happy}}}

Chapter 4 revisits the movement theory of Control, as Chomsky argues in the MERGE framework for an alternative derivation for Control structures which, though based partly on Hornstein (1999), avoids violating Duality₂.

Other approaches have been suggested which involve IM to a theta-position. For instance, Jeong (2006) analyses ditransitives such as *give*, *send*, and *buy* with a low applicative structure, following Pytkäinen (2002). Jeong argues that the semantics of these constructions is best explained through movement to a theta-position,

parallelling Hornstein's (1999) analysis of Control: (30) involves movement of *a book* from one theta-position to another.

(30) $\{_{VP} \text{ John } \{_V \{_{VP} \text{ a book}_i \{_V \{_{LAppIP} \text{ IO } \{_{LAppI} \text{ a book}_i\}\}\}\}\}\}$ (Jeong, 2006:206)

As this approach is analogous to Hornstein (1999), it can easily be adapted in the same way, thus avoiding a Duality₂ violation.²⁹

3.3.2 EM of non-arguments

The more pressing issue with this version of Duality₂ is the question of non-arguments. Arguments EM'ed into a non-theta-position are disallowed, as discussed in section 3.3 and further in section 3.4.1; however, it is clear that many elements EM'ed into the structure are not arguments and are not EM'ed in theta-positions. Obvious examples include adjuncts and heads. The problem may be avoided for adjuncts if these are formed by Pair-Merge rather than Merge, although this is debatable.

However, Duality₂ is meant to derive the copy/repetition and Control/Raising distinctions, as discussed in the following chapter. Therefore, it is problematic that it cannot distinguish copies from repetitions in the case of adjuncts and heads³⁰, e.g., the distinction between (31) and (32).

(31) On Mondays I think that on Mondays he drinks (Collins & Groat, 2018:5)

(32) On Mondays, I think that he drinks

²⁹ Jeong (2006) argues that the analysis can be extended to serial verb and resultative constructions. Interestingly, these can also be analysed in the same way as Control under MERGE (see section 6.6.5 for serial verb constructions and Saito (2022:168) for resultatives).

³⁰ If head movement pertains to externalisation, as Chomsky suggests, then this problem is avoided for heads: the fact that heads are EM'ed in non-theta-positions does not pose any problems for the copy/repetition distinction, since there are no copies of heads in the syntax.

Chomsky et al. (2023:37) mention in a footnote that ‘Duality is relevant only when the merger of arguments (NP and embedded CP) is involved, where it requires EM of an argument to saturate a theta position’. Given the large focus placed on Duality₂ in MERGE, it is perplexing that observation of its apparent restriction to arguments is only mentioned in a footnote rather than being discussed more explicitly.

3.4 Restricting Duality₂ to the A-system

Perhaps Chomsky’s (2021) definition of Duality₂ restricts it to the A-system for the reasons discussed in the previous section. This paper assumes a clear A/A’ separation; for instance, FormCopy (see chapter 4) from an A-position only searches A-positions (Chomsky, 2021:28). Chomsky argues that A-positions yield core semantics, i.e., theta-positions (once again without mention of Case), whereas A’-positions yield discourse- and information-related meanings. Similarly to Chomsky (2007, 2008), this is considered ‘the basic idea that underlies Duality of Semantics’ (Chomsky, 2021:29). The A/A’ distinction and the EM/IM distinction inherent to Duality are once again conflated.

Duality₂ is reformulated as follows:

(33) For A-positions, EM and EM alone fills a θ -position. (Chomsky, 2021:30)

In other words, Duality₂ applies only within the A-system, presumably based on the theta-role/‘everything else’ definition (with no reference to discourse, that being the domain of A’-movement). This big shift from the original formulation of Duality₂ is made without convincing justification.

The status of the T-domain and Case is worth considering here (see also section 2.4). Indeed, though (33) does not bar A-movement, the T-domain is not discussed. Given that Case and theta-positions are tightly connected in past work (e.g.,

Chomsky, 1981; 1982) since both belong to the A-domain, it is curious that theta-roles are given such importance in MERGE while Case is an afterthought, if it is acknowledged at all.

Chomsky (2024), essentially a rephrased version of Chomsky (2021), incorporates Case-positions somewhat more effectively in its new definition of Duality₂.

(34) a. For A-positions, EM is one-one associated with theta position, IM is
one-one associated with argument position

b. IM alone yields A-bar positions. (Chomsky, 2024:35)

This definition is the first to introduce a three-way distinction: theta-positions, argument³¹ (Case) positions, and A'-positions, as opposed to the two-way distinction in most versions of Duality. As such, it appears to be a clarification of (33) to account for A-movement.

(33) and (34) avoid the problems of (26) concerning merger of non-arguments, but the issues concerning the relevance of the A/A' distinction from section 2.4.4 remain. Note also that (33) (but not (34)) allows EM of arguments to non-theta-positions. It is unclear why Chomsky would define Duality₂ in a way that permits this.

Some recent work (e.g., Rugna & Manzini, 2023) uses (33), as it was the most up-to-date version of Duality at the time. Here, EM of arguments to non-theta-positions occurs in certain cases. It is efficient computation that forces IM, rather than Duality₂: IM being simpler than EM, non-theta-positions are filled by IM unless an independent constraint prevents it. As discussed in section 1.1.2, however, efficient computation is no longer relevant when distinguishing Merge and Move, as they are the same

³¹ The use of the term 'argument position' to refer solely to Case positions is unusual, however. In Chomsky's past work (e.g., Chomsky, 1981; 1982), 'argument positions' refer to both theta- and Case-positions.

operation.³² As I discuss in the following section and in chapter 5, theoretical problems arise when allowing merger of arguments in non-theta-positions.

3.4.1 Left dislocation

The theoretical consequences of (33) aside, it is worth considering structures where arguments are base-generated in the left periphery. These present challenges for Duality₁ (discourse realised by EM) and for Duality₂ as in (26) (EM to non-theta-positions), yet are unproblematic for (33).

One such example is left dislocation, whereby an element is EM'ed in the left periphery and in some way linked to a resumptive pronoun in the canonical position (e.g., Alexiadou, 2006). For instance, (35) and (36) both introduce topic meanings, but (36) involves movement to the left periphery.³³

- | | |
|---|--------------------|
| (35) This story, she really liked it | - left dislocation |
| (36) This story, she really liked this story | - topicalisation |

Evidence for (35) involving base-generation rather than movement include a lack of island effects (37) and lack of reconstruction (38).

- (37) a. As for this story, I know the woman who wrote it
b. *This story, I know the woman who wrote
- (38) a. Herself₁, Anna₁ thinks that Susan likes her₁/*herself₁
b. Herself₁, Anna₁ thinks that Susan likes herself₄

³² The aim is to constrain Merge in some way to achieve the desired outcome. With a strict version of Duality₂, Merge can apply more freely, while the weaker version in (33) requires stricter economy constraints.

³³ Interestingly, Chomsky refers to topicalisation and left dislocation in Chomsky (2019a:278), which is surprising given that these are counterevidence for Duality as defined at the time.

How left-dislocated elements are to receive a theta-role is unclear. Presumably, there is a link between the left-dislocated element and the resumptive pronoun which permits this. For further discussion, see section 5.2.4.

3.4.2 Extending (33) to the A'-system

Chomsky (2021) suggests that (33) can be extended to the A'-system if the concept of theta-roles is generalised to include positions in the left periphery, as in Chomsky (2005) (see section 2.3). Thus, another definition of Duality₂ is (39).

(39) For A'-positions, EM and EM alone fills a 'theta'-position.

Aside from the fact that Chomsky's stance on the cartographic tradition varies in his recent work,³⁴ Chomsky does not elaborate on what a left periphery 'theta'-role would be, and this is unclear. Here, I assume that he is referring to the specifiers of heads like Foc or Top. Hayashi (2022:112) discusses a similar idea (although in the context of Duality as in (26)), whereby DPs which are EM'ed with a clause receive a semantic role TOPIC, which behaves like a theta-role. Hayashi argues that phenomena such as left dislocation are unproblematic for Duality as they effectively receive a theta-role.

If we assume (39), are there instances where this version of Duality₂ is applicable? If we propose that Foc and Top heads project 'theta'-positions, might there be evidence barring IM to a 'theta'-position? For instance, if elements base-generated in Spec-Fin are allowed to move to, say, Spec-Top, this would be a counterargument for this version of Duality₂. Further research is required to ascertain whether this is possible.

³⁴ Chomsky et al. (2019:237) criticise it, whereas Chomsky (2024:29) uses it to justify Duality₂.

Thus, Duality₂ in the A'-system seems unclear and potentially flawed. It is also uncertain how this strict division of Duality₂ into an A-system and A'-system version would interact in cases involving A-to-A' movement, e.g., the movement from object position to topic position in topicalisation examples such as (36); the definition in (34) appears preferable in this regard.

In the remainder of this thesis, Duality₂ will refer to (26) (EM to theta-positions, IM to non-theta-positions), as my theoretical critique focusses on Chomsky et al. (2023), which does not refer to the A/A' distinction when discussing Duality₂ (it is actually considered to be eliminable in this work). Nevertheless, given the issues concerning EM of non-arguments, it is important to remember that Duality₂ refers solely to arguments.

3.5 Restricting Duality₂ to the vP-phase

One additional point to consider is whether we can address many of the empirical issues by restricting Duality₂ to the vP-phase. Section 2.4.3 showed that the notion of vP being the thematic phase and CP the scope/discourse phase is too simplistic, as the vP-phase also involves discourse movement. If Duality₂ is restricted to vP, it produces the desired outcome: EM yields theta-positions and IM yields non-theta-positions, including discourse positions in the low focus projection. This is an alternative way of narrowing down Duality₂, as attempted in Chomsky (2021), but without the issues arising from Chomsky ignoring the T-domain or from associating it with the A/A' distinction. Restricting Duality₂ to the vP-phase seems logical, given that vP is where theta-roles are assigned, making it the phase where the question is most pertinent.

3.6 How can Duality₂ constrain Merge?

In section 1.1.5 I introduced a theme of this thesis: the implications for the interfaces of simplifying Merge by relegating elements to the semantics, leading to inconsistencies about whether certain elements are syntactic or interpretive. This issue arises in the context of Duality₂. For Chomsky et al. (2023), Preservation and theta-theory derive Duality₂, which constrains Merge. However, both theta-theory and Preservation pertain to interpretation, not syntax, so it is unclear how they are to constrain Merge (for details on Preservation, see chapter 5).

Indeed, for Chomsky (e.g., Chomsky, 1995b:27), theta-roles are assigned configurationally, rather than being features as in Hornstein (1999:78 et seq.). Theta-roles are computed at interpretation by considering the position in which the arguments are merged. As such, theta-roles, being semantic in nature, are not assigned in the syntax (theta-role assignment in the syntax would violate Preservation, for instance). This is obscured by Chomsky's use of the term 'theta-marking' (Chomsky, 2021:27), which I argue is due to his wish for theta-roles to be relevant as constraints for the syntax.

Indeed, if theta-roles pertain to interpretation, it is unclear how Duality₂ can constrain Merge.³⁵ ³⁶ Chomsky might be stating that Duality applies to the output of Merge as a filter, but the wording is unclear. For instance, Chomsky et al. (2023) are inconsistent in this regard, stating that 'Merge [...] will be constrained by the

³⁵ This is also noted by Mizuguchi (2023:55).

³⁶ This issue appeared as early as Chomsky (2000), where the Theta-Criterion was partly a derivational constraint (only arguments can be merged in theta-positions, and only non-arguments in non-theta-positions). Although the Theta-Criterion (now referred to as Univocality) is now considered a filter (see section 4.3), the ambiguity as to the status of theta-roles in the syntax is not new (note also that the fact that some elements of theta-theory are considered filters (Univocality) and others constraints (Duality) is also unconventional).

(language specific) requirements of interpretation, including [theta-theory].’ This is followed by a footnote, clarifying that ‘Merge itself does not know what will be the interpretation of an object it builds. However, the interpretative component can act as a ‘filter’ on the output of Merge, and hence constrain its operation.’ (Chomsky et al., 2023:31). This illustrates that Chomsky et al. are reluctant to commit, aiming for Duality to constrain Merge despite its status as an interface condition. Their discussion strongly implies that it acts as a constraint rather than a filter, however: on page 26, for instance, they state that ‘Duality of Semantics (like RR³⁷) constrains how Merge applies, thereby contributing to the reduction of computational complexity’. If Duality acts as a filter, it cannot reduce computational complexity.

3.7 Summary

The duality of semantics has become much more specific and relevant to the theory in recent years. In MERGE, it is a principle derived from Preservation and theta-theory which constrains Merge. The predictions of Duality₁ and Duality₂ overlap but are not identical.

The main variations of Duality₂ are summarised here:

- (40) EM to theta-positions, IM to non-theta-positions. (= (26))
- (41) For A-positions, EM and EM alone fills a theta-position, and
 For A'-positions, EM and EM alone fills a ‘theta’-position. (= (33) and (39))
- (42) For A-positions, EM is one-one associated with theta-position, IM is one-one
 associated with argument-position. IM alone yields A-bar positions. (= (34))
- (43) At the vP-phase, EM to theta-positions, IM to non-theta-positions.

³⁷ Resource Restriction

(40) implies that EM must always be to a theta-position, which I have argued is problematic given that not all elements entering the derivation will receive theta-roles, for instance heads and adjuncts. The definition in (41) avoids this issue but inherits concerns about the A/A' distinction and the notion of A'-theta-roles is unclear. Additionally, the status of the T-domain is uncertain, with (42) being preferable in this regard, providing a three-way distinction mirroring standard clausal structure. (41) also struggles to account for A-to-A' movement, unlike (42). One possibility to the array of issues surrounding Duality₂ might be to restrict it to the vP-phase as in (43). Nevertheless, a central issue remains: given that theta-roles are assigned configurationally and are not present in the syntax, it is unclear how Duality₂ can constrain Merge.

Chomsky claims that Duality₂ as in (40) derives the copy/repetition and Raising/Control distinctions. Chapter 4 evaluates these critically. Henceforth, unless mentioned otherwise, Duality₂ refers to this definition as it is the most relevant for my discussion.

4 Duality₂ and the copy/repetition/PRO distinction

The previous chapter introduced Duality₂, the version of the duality of semantics used in MERGE. Chomsky argues that Duality₂ derives the copy/repetition distinction and the Control/Raising distinction. This chapter critically examines both proposals from an empirical and theoretical standpoint.

4.1 Distinguishing copies and repetitions

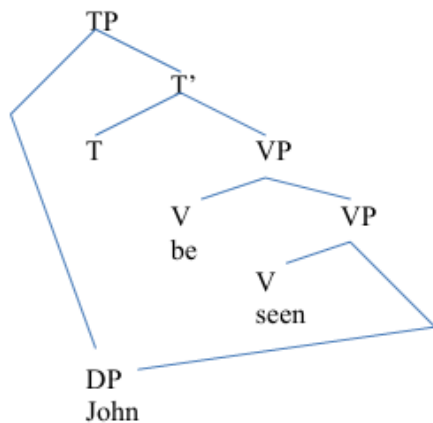
The copy/repetition distinction is a crucial concern in minimalism given the copy theory of movement (Chomsky, 1995b). Since movement does not create a trace, instead leaving a copy in the original position, there is no obvious distinction in the structure between copies and repetitions, i.e., identical LIs that were independently merged. Various approaches have been used to make this distinction. In their overview, Collins & Groat (2018) argue that existing approaches are either problematic or stipulative. This section summarises those most relevant for my discussion.³⁸

4.1.1 Multidominance (Citko, 2005; 2011)

Multidominance produces a structure in which, rather than an additional copy being generated through movement, a single LI occupies two positions simultaneously. This is illustrated in (44).

³⁸ For space reasons, I do not discuss other issues with copy theory, including the different interpretations given to A- and A'-copies in operator-variable approaches (see also footnote 21).

(44)



(Collins & Groat, 2018:2)

Multidominance attempts to avoid problems which arise when creating copies (for discussion of issues surrounding copies, see chapter 5). However, Collins and Groat argue that this approach is ineffective, since Merge creates sets rather than trees. Indeed, (44), when converted into a set, yields (45). Distinguishing the sets for *John was seen* (46) and *John saw John* (47) requires indices, which violates the Inclusiveness Condition.³⁹

(45) {John, {T, {be, {seen, John}}}}

(46) {John₁, {T, {be, {seen, John₁}}}}

(47) {John₁, {T, {see, John₂}}}

(Collins & Groat, 2018:3)

Collins and Groat's analysis abstracts away from the specific details of the structures involved, but this appears problematic. Indeed, these two structures are also distinguished by the featural make-up of the verbal heads that are postulated to distinguish active and passive argument structure.

³⁹ Indices used in this thesis serve solely for the purpose of readability and do not have any theoretical significance.

Nevertheless, Multidominance remains problematic under MERGE. Indeed, Citko (2005) derives Multidominance structures from Parallel Merge, which is disallowed under MERGE (see section 1.1.3).

4.1.2 Chains

Chains were used to derive the copy/repetition distinction in Chomsky's early minimalist work. *John was seen* and *John saw John* are distinguished by the number of chains involved, without the need for indices:

- (48) *John was seen* involves one chain: <first occurrence of *John*, second occurrence of *John*>
- (49) *John saw John* involves two chains: <first occurrence of *John*>, <second occurrence of *John*> (Collins & Groat, 2018:4)

However, this approach requires definitions of both chains and occurrences and assumes that chains are formed as a part of IM, greatly complicating the definition of Merge. Lastly, Transfer must refer to chains rather than syntactic objects.

4.1.3 Phase-level memory

Consequently, Chomsky transitioned from chain-based approaches to using phasal memory. Two versions have been proposed. The first, introduced in Chomsky (2008) (which is chronologically prior to Chomsky (2007) in the development of Chomsky's thought), refers to lexical selection. Chomsky (2008:145) suggests that 'within a phase each selection of an LI from the lexicon is a distinct item, so that all relevant identical items are copies.' In other words, at the phase level, the system remembers how many times a lexical item was selected from the lexicon. Thus, *John was* selected once in (46) and twice in (47), so the inscriptions of *John* are copies in (46) and repetitions in (47).

However, it is unclear how selection from the lexicon is recorded in memory. Collins and Groat suggest referring to the sequence of workspaces. However, this is problematic for reasons discussed below, including uncertainty about how the information concerning lexical selection is sent to the interfaces. Lastly, they argue that this approach struggles to deal with complex syntactic objects. Indeed, in (50), the copy/repetition distinction between the two instances of *the governor of California* cannot be made by referring to the number of times each LI was selected from the lexicon.

(50) The governor of California was seen ~~the governor of California~~

Chomsky's second approach suggests that at the phase level, the system remembers if IM or EM has applied, and 'all and only repetitions formed by IM within a phase are copies' (Chomsky, 2007:10). As with the version from Chomsky (2008), Collins and Groat argue that it is unclear how this information is stored, unless reference is made to sequences of workspaces. Sending the interfaces a sequence of workspaces rather than syntactic objects is undesirable, undermining the concept of a workspace, and this approach appears stipulative. The notion of a sequence of workspaces is used solely to draw the copy/repetition distinction and does not emerge naturally from this approach. Note also that the notion of a sequence of workspaces might not be possible in MERGE since derivations are Markovian (see section 1.1.4).

4.2 Copies in the UCLA Lectures

In the UCLA Lectures (Chomsky, 2019b), Chomsky moves away from phase-level memory approaches to copies and repetitions; the distinction is made using Duality₂. The definition of Duality₂ is repeated in (51).

(51) EM to theta-positions, IM to non-theta-positions. (= (40))

At the phase level, the system knows that elements in theta-positions are repetitions (i.e., EM'ed there), while those in non-theta-positions are copies (i.e., IM'ed there) (Chomsky, 2019b:44).⁴⁰

Collins & Groat (2018) criticise a similar approach to the copy/repetition distinction, from personal communication with Chomsky, Ott, Blümel, and Seely, before the introduction of Duality₂:

(52) If DP₁ is in a theta-position [...], then DP₁ and DP₂ must be repetitions.

Otherwise they are copies.⁴¹ (Collins & Groat, 2018:5)

Collins and Groat argue that since theta-roles are semantic notions, the phonological component should be unaware of them. Thus, a theta-role-based distinction fails to account for the close interdependence between the phonological and semantic consequences of the copy/repetition distinction. In his Duality-based approach, Chomsky may consider theta-roles to be assigned at Transfer, thus reaching both interfaces, but this is not made explicit. These concerns resemble the issue discussed in section 3.6: the impossibility for Duality, a semantic notion, to constrain Merge.

Collins and Groat also emphasise that this approach to the copy/repetition distinction cannot deal with non-arguments. This also applies to the Duality-based distinction (see section 3.3.2), leaving open the question of adjunct movement, for instance.

⁴⁰ Chomsky's definition here has an exception: an element in a theta-position is a repetition 'unless it's been raised, in which case it's a copy of what's been raised' (Chomsky, 2019:44). This stipulation does not appear in later works. Its rationale is unclear, likely arising from Chomsky's tendency to overlook the T-domain in the context of Duality, as discussed.

⁴¹ See also Hayashi (2022) for a similar approach, in the framework of Chomsky (2021).

Lastly, Collins and Groat note that this approach allows copies to be formed by EM.

Interestingly, it is notable that were such an approach plausible, it would make the distinction between External and Internal Merge irrelevant for the determination of the copy/repetition distinction; all structure-building could be External Merge, with the distinction falling out of the semantic interpretation of theta-roles. (Collins & Groat, 2018:5)

Collins and Groat highlight that this might have problematic consequences. The Duality-based distinction makes different predictions; indeed, not all structure-building can be EM as this would violate Duality₂ (Duality is used to justify the existence of both types of Merge, see section 3.1). However, this quote is interesting given Chomsky's later introduction of a copy formation operation, which allows for precisely this: copies formed by EM.

4.3 FormCopy (FC) and the Raising/Control distinction

Until Chomsky (2021), Chomsky assumed that copies are a byproduct of IM. Chomsky (2021) changes this by introducing the notion of FormCopy (FC), a third-factor⁴² interpretive rule, subject to the PIC, which can assign the copy relation to identical inscriptions at the phase level. FC finds copies using the freely available operation of MS. FC, in (53), makes explicit the implicit copy formation operation existing in earlier work.

(53) Where X, Y are structurally identical, FC(X, Y) interprets X, Y as copies, that is, the inscriptions are interpreted in exactly the same way.

⁴² Chomsky claims that FC is a third-factor rule, presumably since it depends on MS. MERGE aims to simplify the syntax by relegating as many elements as possible to the interfaces or the third factor, but this is only valid if it can be properly justified, which is not always the case. Indeed, it is unclear how FC is a principle of efficient computation or general law of nature.

(Chomsky et al., 2023:24)

FC applies freely, but its output is constrained by the Univocality of theta-theory (i.e., the Theta-Criterion in past work) as in (54) (= (27))

- (54) every theta role must be assigned to one, but only one argument; and every argument must receive one, but only one theta role.

(Chomsky et al., 2023:31)

Indeed, application of FC in (55), yielding (56), is barred by Univocality.^{43 44}

- (55) John saw John

- (56) *John saw ~~John~~

FC, Duality₂, and the Markovian property of derivations are relevant for distinguishing between Raising (58) and Obligatory Control (57) structures. Chomsky (2021:21-25) considers insights from Hornstein's (1999) Raising analysis of Control (see section 3.3.1), arguing that the two structures are similar but not identical. The Markovian property of derivations entails that the system does not know if identical inscriptions were formed by EM or by IM.

- (57) The cat wants ~~the cat~~ to eat a fish (= (28)) – Control

{_{VP} **the cat** {_{VP} wants {the cat to eat a fish}}}⁴⁵

- (58) The cat seems ~~the cat~~ to be happy (= (29)) – Raising

⁴³ A similar idea is mentioned in Epstein et al. (2015:218) in a footnote:

A less stipulative algorithm would be this: Freely assume <X,X> to be copies or assume them to be two separate arguments. This would arguably suffice empirically; that is, if *He was arrested t(he)* is taken to contain two arguments, there is a theta/Full Interpretation/gibberish problem at CI. If *He thinks he is smart* is taken to involve two copies created by IM, then, again, there is a theta/Full Interpretation/gibberish problem at CI.

⁴⁴ Presumably, (56) is also barred by the two occurrences of *John* bearing different Case features and thus being non-identical. The fact that this is never mentioned by Chomsky once again highlights the fact that, in MERGE, Case appears to be an afterthought.

⁴⁵ The structure obviously continues past vP but is illustrated as such for ease of exposition.

{_{VP} **the cat** {_{VP} seems {the cat to be happy}}}

Due to Duality₂, the bolded inscription of *the cat* in (57) cannot be formed by movement, as it is in a theta-position, whereas the bolded inscription in (58) must be formed by movement as it is in a non-theta-position. However, the lower inscription in (57) is unpronounced: FC, being Markovian, cannot tell whether inscriptions have been EM'ed or IM'ed, and can apply in (57), forming a copy relation between the two inscriptions. Such copies are called M-gaps ('Markovian-gaps'; or copy-with-EM in Chomsky et al. (2023) (distinguished from copy-with-IM⁴⁶)) and derive Control structures without using PRO. For Chomsky, this illustrates the enabling function of the Strong Minimalist Thesis (SMT), arguing that Obligatory Control arises naturally from his definition of MERGE without requiring additional explanation.⁴⁷

This approach is desirable in that it reduces the theoretical postulates; Control and Raising are distinguished by the nature of their derivations. However, various concerns arise. Firstly, the existence of FC is problematic. Chomsky has stated in previous work that a copy formation rule is unnecessary (Chomsky, 2007:10; Chomsky et al., 2019:246).⁴⁸ Secondly, the description of FC is ambiguous. In later work, FC clearly also applies in cases of IM, but in Chomsky (2021), FC is sometimes described as IM without language-specific conditions (e.g., p.21). This

⁴⁶ The terminology is confusing as 'copies' are both what is produced by Merge in the new formulation of Merge, and what is assigned by FC. See section 5.5.

⁴⁷ This approach is a return to Equi NP Deletion (Postal, 1970; Chomsky, 1981), which involved deletion of the subject of the embedded clause when it was identical with the subject or object of the main clause. Equi NP Deletion was abandoned in favour of PRO-based accounts for reasons discussed in the following section, as the empirical issues with the FC-theory of Control inherit the problems of Equi NP Deletion.

⁴⁸ Admittedly, what was referred to as a copy formation rule in these works differs somewhat (although not entirely) from FC.

appears to be mainly an issue of poor exposition. See chapter 5 for further issues concerning FC.

FC-theory invalidates the copy/repetition distinction from Chomsky (2019b). It is no longer possible to assert that elements in theta-positions are necessarily repetitions. FC can apply in such structures, creating a copy, as in Control derivations. The copy/repetition distinction now reduces to whether FC has applied. While simple, this is not particularly explanatory. This approach also struggles to account for the copy/repetition distinction for non-arguments. Indeed, as discussed, the output of FC is constrained by the Univocality of theta-theory, which is not relevant for heads, adjuncts or elements base-generated in the left periphery, leaving them largely unaccounted for.

4.4 Empirical issues

This section summarises key empirical issues facing the FC approach to Obligatory Control (OC), primarily drawing on Landau (2024) which provides a more in-depth analysis.⁴⁹ The key point is that for Chomsky, the differences between Control and Raising structures are differences in interpretation, whereas for Landau, these distinctions are inherently syntactic. As discussed, for Chomsky, IM-copies (as in Raising) and EM-copies (as in Control) are indistinguishable in the syntax due to the Markovian nature of derivations. Thus, syntactic differences between Raising and Control structures cannot be accounted for. Although Chomsky mentions many of the aspects discussed by Landau, he does not explore them in detail, and in-depth analysis reveals limitations to an interpretive approach.

⁴⁹ Many of the empirical issues concerning the FC-theory of Control are the same as those faced by the movement theory of Control due to the similarities between the derivations.

Landau (2024:5) summarises the properties of OC structures according to FC-theory as follows:

- (59) a. The controllee is a null full NP, which is structurally identical to the controller
b. The Control complement is not a phase.

In theories of Control such as Landau's Two-Tiered Theory of Control (TTC) (e.g., Landau, 2015; 2024), the controllee is a pronoun, and the Control complement is unproblematically a phase. I now discuss criticisms of (59).

4.4.1 Absence of reconstruction effects

The first issue with FC-theory is the absence of reconstruction effects in OC, unlike in Raising. This is illustrated in (60)-(61), where (60) allows specific and numerical readings⁵⁰ and (61) allows only the specific reading.

- (60) How many students were expected to fail the exam? – Raising
(61) How many students expected to fail the exam? – Control

(Landau, 2024:8)

The same effect can be seen with weak-crossover violations, with violations repaired in the Raising configuration but not in the Control structure:

- (62) One interpreter each seems [t to have been assigned t to the diplomats]
– Raising
(63) *One interpreter each tried [PRO to be assigned t to the diplomats]
– Control

⁵⁰ The specific reading is 'For what n: there are n-many students x such that it was expected that x would fail the exam'; the numerical reading is 'For what n: it was expected that there would be n-many students x such that x would fail the exam'.

(Landau, 2024:9, citing Chomsky et al. (2023:45) (citing Burzio (1981, 1986)))

Since Chomsky claims that Raising and Control constructions have the same structures as far as the syntax is concerned, the reason for reconstruction effects in Raising but not in Control is unclear. Nevertheless, Chomsky cites such examples, arguing that the differences between the two structures are interpretive.

INT⁵¹ cannot distinguish them, but other operations can, and do. [...] The distinction between the two kinds of copy seems well established from several perspectives. It therefore provides empirical support for the assumption that the Duality of Semantics principle [(22)], on which the distinction rests, is indeed an LSC. (Chomsky, 2021:22).

Chomsky uses this as evidence for Duality, i.e., the necessity for both EM- and IM-copies to account for this difference (rather than using an approach like (52)).

How do EM- and IM-copies account for this difference? Chomsky et al. (2023) elaborate, arguing that Preservation can distinguish them.⁵²

This distinction arises not because of a distinction between PRO and trace, but because of the difference in the derivation of the two structures: [(62)] involves IM of *one interpreter* from the positions marked *t* and reconstructed in that position, while in [(63)], there are two separate instances of *one interpreter* (as we have just detailed) given Preservation, and there is no reconstruction; rather, interpretation of an independent element, and thus

⁵¹ Here, INT refers to ‘an expository device to lay out the options for the next step of derivation.’ (Chomsky, 2022:361), i.e., to the syntax; unlike in some other work such as Manzini & Roussou (2024a), where it refers to interpretation.

⁵² Chomsky et al. (2023:44) also suggest that the trace/PRO distinction can be achieved by theta-theory – essentially Duality₂ (see section 5.2.1), though this has the same issue, since theta-roles are interpretive.

each is stranded. [...] The distinction follows from structurally identical inscriptions being derived from IM, in which case in the eyes of Preservation they are in fact one and the same inscription, as opposed to the structurally identical inscriptions being construed from separate instances of EM.

(Chomsky et al., 2023:45)

However, Preservation pertains to interpretation, syntactic operations being Markovian. IM- and EM-copies cannot be distinguished in the syntax, yet the distinction between (62) and (63) is syntactic, not semantic. Thus, while Chomsky interprets this data as supporting FC-theory, it actually contradicts his claims. For Landau, however, such data follows automatically from the absence of a copy of the controller in the lower clause.

Chomsky and Landau both cite *en*-cliticisation in French. In Raising structures, the partitive clitic *en* can be stranded in the embedded clause (64), which is impossible in Control structures (65).

(64) a. Le directeur du département semble être accepté
the head of.the department seems to.be accepted
‘The head of the department seems to be accepted’ – Raising

b. Le directeur semble en être accepté
the head seems of.it to.be accepted
[Le directeur en]_i semble [[le directeur en]]_i en être accepté [le directeur en]_i
‘The head of it seems to be accepted’

(65) a. Le directeur du département espère être accepté
the head of.the department hopes to.be accepted
‘The head of the department hopes to be accepted’ – Control

b. *Le directeur espère en être accepté
 the head hopes of.it to.be accepted

*Le directeur_i espère [PRO_i en être accepté PRO_i]

‘The head of it hopes to be accepted’ (Landau, 2024:10)

For Landau, this follows from the absence of a lower copy in (65), where the clitic can be stranded. The clitic is a remnant, with cliticisation preceding Raising. PRO, however, cannot host a copy of the clitic.

Chomsky (2021:22) cites *en*-cliticisation without elaboration. The differences between (64) and (65) and are clearly syntactic rather than interpretive. Landau argues that such data is particularly problematic, as look-ahead is required to allow the derivation of (64) but not (65) in Chomsky’s approach.

At the point of *en*-cliticisation [...] the Raising derivation and the Control derivation have constructed exactly the same object: [_{InflP} Infl [_{AuxP} être [_{VP} accepté [_{DP} le directeur en]]]]. There is simply no way to allow, as the next step, Merge (*en*, InflP) *only* if two steps downstream a raising predicate (*semble*) is to be merged, and block it if a control predicate (*espère*) is. This kind of look-ahead is unavailable, and in any way would be arbitrary. (Landau, 2024:11)

Thus, the syntactic differences between Control and PRO structures cannot be accounted for by FC-theory, as it considers them indistinguishable as far as the syntax is concerned.

4.4.2 Control structures include a CP-phase

Control structures are standardly analysed as containing a CP-phase (Landau, 2015). This is problematic for the FC-theory of Control, as FC is subject to the PIC. If

Control structures contain a CP-phase, it is unclear how the lower copy can be related by FC to the higher one⁵³.

Landau provides evidence for the existence of a CP-phase in Control structures, and a portion of this evidence is discussed here.⁵⁴ Firstly, interrogative infinitives show wh-island effects, i.e., evidence for a phase boundary, e.g., in cases of adjunct extraction in (66)-(67).

(66) *How_i did Kevin ask [what PRO to clean t_i]?

(67) How_i did Kevin offer [PRO to clean the bath t_i]? (Landau, 2024:17)

Landau also argues that, although there might be cases of Control where a CP boundary is absent, the presence of even a single instance with a CP-phase poses a challenge to FC-theory. Propositional complements, which Wurmbrand & Lohninger (2023) argue are the largest clausal complements (so definitely containing a CP!), are possible in OC environments, as shown in (68)-(70).

(68) Mary pretended/claimed to be busy with customers

(69) Jane regretted/hated watching *A Nightmare on Elm Street*

(70) Er glaubt, gewonnen zu haben
he believes won to have

‘He believes himself to have won’ (Landau, 2024:18)

⁵³ Specifically, FC is subject to the PIC₁, the version used in MERGE (see footnote 24). The PIC₂, a less strict form of the PIC, states that the complement of the phase head HP is inaccessible to operations at the level of the next phase ZP, while the head H and edge remain visible (e.g., Citko, 2014:33). If FC were subject to the PIC₂, the problem concerning control remains, despite the PIC₂ being less strict. Indeed, the phase head complement becomes inaccessible upon merger of the next phase head. In a structure such as (2), the TP complement of C becomes inaccessible upon merger of v *wants*, so when DP *she* is merged, the lower DP *she* is inaccessible and FC cannot apply.

(2) {{DP she} {VP wants {VP ~~wants~~ {CP C {TP {DP ~~she~~} {TP to be happy}}}}}}

⁵⁴ Further evidence includes data from languages which use the same complementiser for finite clauses and infinitival OC complements (e.g., Swedish), and the edge of OC complements being a site of Q-stranding in Ulster English long-distance wh-movement.

The first response when considering such data in the context of FC-theory is to suggest movement of the controllee to Spec-CP.⁵⁵ In a footnote, Landau argues that this is problematic for wh-infinitives, but does not discuss in depth. Another potential issue with such movement is that it instantiates improper movement, i.e., movement from an A'-position to an A-position. This is standardly disallowed (Williams, 2003:ch3). Nevertheless, I assume movement to Spec-CP when illustrating Control structures under FC (see chapter 6) as it is the least undesirable option.⁵⁶

4.4.3 Further issues with the FC-theory of Control

This section briefly outlines further empirical issues with the FC-theory of Control. As FC-theory is relatively new, many Control-related phenomena have not yet been fully addressed. These issues may be resolved in the future and, indeed, some of the works I cite are manuscripts, reflecting the ongoing, cutting-edge nature of the research.

As discussed in section 4.4.1, FC-theory cannot explain the evidence for PRO being a pronoun rather than an NP copy. Landau argues that this is a fundamental aspect of Control. He notes that FC-theory allows overgeneration of structures where the controlled copy in (71)-(72) is produced by movement. These structures cannot be ruled out by Duality₂ as the movement is to a non-theta-position (given that the structure is a passive).

(71) *John was hoped ~~John~~ to win the game

⁵⁵ An alternative solution is to redefine the PIC as done by Saito (2022).

⁵⁶ Jakielaszek (2023) suggests using feature inheritance to solve this issue. Feature inheritance is a concept from Chomsky (2007, 2008) which has various issues I cannot discuss for space reasons. Interestingly, Chomsky has never discussed feature inheritance in MERGE.

(72) *The committee was regretted/hated ~~the committee~~ to have passed the
decision (Landau, 2024:22)

Additionally, Landau argues that FC-theory cannot explain why Control is obligatory in certain cases. Indeed, FC applies freely, and will overgenerate structures like (73).

(73) *John tried John/Mary to win

Chomsky et al. (2023:43) argue that such examples involve ‘SM-blocked but CI-convergent derivations’, which presumably refers to grammatical sentences which pose a problem for externalisation. I discuss this highly problematic notion in section 5.4.

Lastly, FC-theory presented so far struggles to account for partial⁵⁷ (74), split (75) and Non-Obligatory Control (76)-(77). In Partial and Split Control, the controllee is not identical to the controller so FC cannot have applied. In Non-Obligatory Control, there is either no antecedent (76) or an antecedent which cannot be related to the controllee by FC for structural reasons (locality restrictions, absence of c-command) (77).

(74) Susan persuaded the chair to meet without her

(75) Susan persuaded Dan to continue working together on this
(Manzini & Roussou, 2024a:13)

(76) It suffices to win

(77) To win would bother John (Manzini & Roussou, 2024a:7)

Manzini & Roussou (2024a; 2024b) sketch a tentative solution to these types of Control in FC-theory, based on the idea that deletion is ‘No Transfer’ (Saab, 2009).

⁵⁷ Chomsky (2021) outlines a solution to Partial Control which Landau (2024) criticises; I refer the reader to these works for further discussion.

No Transfer can be licensed by FC (deriving OC) and Agree (deriving Finite Control),⁵⁸ and under default feature values (deriving arbitrary Non-Obligatory Control). Logophoric Non-Obligatory, Partial, and Split Control are derived by logophoric binding via Spec-CP. These approaches are tentative, as this is early work, and time is needed for further refinement.

4.5 Summary

This chapter has discussed how Chomsky applies Duality₂ to derive the copy/repetition and the Control/Raising distinctions. I introduced Chomsky's (2021) FC operation and his approach to Control, with a discussion of empirical issues facing FC-theory.

The FC operation significantly alters the status and nature of copies in the theory. Indeed, Chomsky's (2019b) copy/repetition distinction based on Duality₂ is invalidated as, with FC, copy relations no longer come for free with IM. This notion introduces a range of theoretical issues concerning the nature of derivations and copies, which are the focus of the following chapter.

⁵⁸ Roberts (2024) also highlights the similarities between FC and Agree, which may be a promising direction for further research.

5 FormCopy, Preservation, and Markovian derivations

The previous chapter introduced the FC operation and discussed empirical issues facing the FC-theory of Control. This chapter discusses the theoretical implications of FC more generally. With FC, copies no longer come for free with IM, yet I argue that the existence of Preservation undermines this point somewhat. I examine the interactions between FC and Preservation, which both refer to copies and as such complicate the status of copies in the theory. I also discuss questions which arise given that derivations are now considered Markovian (see section 1.1.4).

5.1 Preliminary notions

5.1.1 Preservation and Stability

Before discussing the interaction between FC and Preservation, it is important to clarify the nature of Preservation and its relationship to previous principles introduced in the theory, particularly Stability. Stability is the notion that ‘each occurrence of an expression must have exactly the same interpretation’ (Chomsky, 2021:16-17) and can be traced back at least to Lasnik & Uriagereka (2005:53,122). Preservation was not introduced until Chomsky (2022), and its definition, repeated in (78), resembles that of Stability.

(78) the interpretation of an inscription does not change in the course of computation (Chomsky et al., 2023:22).

A subtle difference exists, however; Stability appears to refer to representations whereas Preservation refers to derivations/computation. This does not make Preservation a derivational operation; rather it is an interpretive principle (and thus not Markovian), albeit one which refers to derivation. Chomsky et al. (2023) mention

this distinction, using the term ‘UIII’ instead of ‘Stability’ – although they appear to be the same.

The standard assumption with formal systems, then, is Uniform Interpretation of Identical Inscriptions (UIII): structurally identical inscriptions are interpreted in exactly the same way. *This is not to be confused with Preservation*: from Preservation it follows that a single inscription can’t change its interpretation in the course of the derivation. From UIII, on the other hand, it follows that for formal systems any set of structurally identical inscriptions in a given ‘Workspace’ are interpreted in the same way. (Chomsky et al., 2023:23, emphasis mine)

Following this quote, Chomsky et al. explain that UIII (= Stability) is derived from the application of FC. Chomsky (2021) considers FC to be subject to Stability. Although space constraints prevent a detailed discussion of the distinction between these two suggestions and associated issues, it is worth noting the emerging potential redundancy between FC and Preservation as mechanisms involved in copy formation, a topic to which I return in section 5.2.

To avoid unnecessary complexity, I henceforth focus solely on Preservation, even in cases where it overlaps with Stability, as it is most relevant to my discussion.

Stability/UIII does not play a significant role in MERGE, and the existence of Stability, alongside Preservation and FC, appears to be redundant, reflecting the gradual development of the theory.

5.1.2 Theta-linking

One final element requires consideration at this point, namely theta-linking. Theta-linking, introduced in Chomsky (2021), states that, supposing τ assigns a theta-role to the structural position $P(\tau)$:

(79) X is θ -linked to $P(\tau)$ if a copy of X occupies $P(\tau)$. (Chomsky, 2021:26)

Thus, theta-linking stipulates that the copy of an element in a theta-position will bear the same theta-role. This rule appears to be necessary because Chomsky construes theta-roles as being assigned configurationally (see section 3.6). If theta-role assignment is configurational, how can higher copies bear the same theta-roles as their lower copies? Theta-linking is an attempt to prevent movement to a non-theta-position from involving ‘loss’ of a theta-role.

For illustration, at the vP-phase level, *which book* in (80) must move to the edge to be able to move further. Transfer renders the greyed-out constituents inaccessible; thus, at the CP-phase level, the theta-role of *which book* is not accessible as the configuration which assigned it has been transferred. Note that the same problem does not appear with agents, as these are base-generated at the edge.^{59 60}

(80) {_{CP} which book {_{CP} did John {_{vP} ~~which book~~ John read ~~which book~~}}}

Through theta-linking, Chomsky wishes to derive the theta-relation between the two copies since FC does not assign theta-roles. However, nothing is derived; instead, a new rule is stipulated. Aside from the fact that theta-linking is an additional complication to the theory which is undesirable by itself, it is unclear why theta-

⁵⁹ Chomsky provides a new definition of theta-theory to account for theta-linking (Chomsky, 2021:27).

⁶⁰ The example Chomsky gives to illustrate the necessity of theta-linking is vISH, which he argues ‘has led to considerable complication in earlier work’ (Chomsky, 2021:26). Since vISH involves a theta-position at the vP edge, accessible at the phase level, it is not entirely clear how theta-linking is relevant here.

linking should be able to see into the phase, other than by necessary stipulation.⁶¹

Furthermore, the definition of ‘copy’ in (79) is unclear. Indeed, if Chomsky takes ‘copy’ here to mean elements to which FC has applied, which is how he defines ‘copy’ in this work, then theta-linking seems to pose a problem for Control structures. Indeed, the two identical inscriptions put in a copy relation by FC would end up being theta-linked. For instance, in (81), the lower inscription of *the cat* is EM’ed in a theta-position. FC forms a copy relation between both inscriptions. Now, the higher copy is theta-linked to the lower one by (79). As such, it now bears two theta-roles, which is disallowed.

(81) The cat wants ~~the cat~~ to eat a fish (= (28)/(57))

{_{VP} **the cat** {_{VP} wants {the cat to eat a fish}}}

Section 5.2.4 returns to this issue in the context of Rugna & Manzini’s (2023) work.

One solution would be to assume that theta-linking refers solely to IM-copies, i.e., the old definition of ‘copy’, which appears to be Chomsky’s approach, although this is never mentioned explicitly. As we will see in the rest of this chapter, reference is still being made in many cases to this old, more ‘fundamental’ version of copies, despite the existence of FC and the Markovian nature of derivations in MERGE. The meaning of the word ‘copy’ becomes notoriously unclear after the appearance of FC; see section 5.5.

Another issue with theta-linking concerns its status as a syntactic operation.

Chomsky views theta-roles as an interpretive property, although, as we have noted, MERGE is inconsistent about this. The same problem exists here. If the purpose of

⁶¹ One might suggest that theta-linking can see into the phase given the interpretive nature of theta-roles, but this is inconsistent with theta-linking being a syntactic operation, to which I return shortly.

theta-linking is to derive the correct theta-roles despite the PIC, theta-linking must be syntactic (the PIC not being relevant at interpretation). However, theta-roles being interpretive, the requirement for theta-linking within the syntax appears contradictory. It is also unclear how copies are to be theta-linked given that assignment is by definition configurational: theta-roles are assigned to specific positions, they are not features which can be copied.

Theta-linking is absent from Chomsky et al. (2023) but reappears in Chomsky (2024). Here, it is stipulated that

(82) A-positions are theta-linked; A-bar positions are non-theta-linked non-arguments. (Chomsky, 2024:28)⁶²

(82) appears to be an attempt to justify the validity of linking Duality₂ to the A/A' distinction, despite the existence of non-theta-positions in the A-system, namely Case-positions (see sections 2.4 and 3.4). Indeed, 'In the system of A-positions, then, Merge always yields a semantic role, either assigning a theta role or a theta-linked argument role' (Chomsky, 2024:29). Thanks to theta-linking, the A-system is always associated with theta-roles, since Case-positions are theta-linked. This, however, appears to be an attempt at justifying an assumption by a stipulation, given the clear stipulative nature of theta-linking.

Thus, theta-linking appears to be an undesirable stipulation, inconsistent with the nature of theta-roles as interpretive properties. Furthermore, I argue that

Preservation makes theta-linking obsolete.⁶³ Indeed, if Preservation states that

⁶² Note that this is a particularly unusual definition of A'-positions. Standardly, arguments can undergo A'-movement; A'-positions are by no means restricted to non-arguments.

⁶³ Although the validity of this depends on our exact definition of 'copy' as Preservation refers to IM-copies alone (see section 5.2). Preservation cannot derive theta-linking as used by Rugna & Manzini (2023) (see section 5.2.4).

copies cannot change in interpretation in a derivation, it is unclear why a linking operation should be necessary to keep track of thematic information. Nevertheless, the precise way in which Preservation enables the higher copy to bear the same theta-role as the lower one despite the configurational nature of theta-role assignment is unclear, which might explain why theta-linking was introduced.

5.2 FormCopy and Preservation

This section examines the interactions between FormCopy and Preservation, two notions both referring to copies. With FC, copies no longer arise automatically through IM; however, the existence of Preservation undermines this idea. Although Preservation is only explicitly discussed in the context of Duality₂, it keeps track of copy relations (but only of ‘copies’ in the traditional sense, as the output of IM⁶⁴) – and is thus relevant to a discussion of copy theory in MERGE.

5.2.1 IM-copies no longer come for free

The existence of both FC and Preservation naturally leads to terminological confusion. Alongside the fact that for so long in the theory copies have been assumed to arise automatically, this renders work that uses FC occasionally inconsistent about whether copy relations come for free. For instance, Saito (2022:172) discusses (83) in the context of remnant movement and MY (see section 5.6.3).

(83) I wonder how likely to win John is

{_{CP} {_{AP} how likely {~~John~~₃ to win}} {_C {_{IP} John₁ {is {_{AP} how likely {~~John~~₂ to win}}}}}}

⁶⁴ Thus, FC applying to form an M-gap does not violate Preservation despite the two copies bearing different theta-roles. Indeed, these copies are not formed by IM, there is no inscription changing in interpretation in the course of a derivation.

Problems arise upon closer inspection, however. If copies no longer come automatically but must be assigned by FC, it is unclear how *John*₃ and *John*₁ can be copies. Indeed, FC requires c-command and is subject to the PIC, both of which are problematic here.⁶⁵

Thus, FC involves a significant conceptual shift concerning copies, which is not always consistently applied in recent papers. The existence of Preservation arguably indicates Chomsky's reluctance to lose the notion that copies come for free with IM. Although the copy theory of movement has its own issues (see section 4.1), FC, with copies no longer derived automatically from IM, presents its own challenges. Specifically, it does not prevent IM from occurring without application of FC (that is, in effect, repetitions being derived from IM), clearly an undesirable outcome. This might be disallowed by Preservation, as an IM'ed element and its 'lower copy', if not put in a copy relation by FC, are not identical; however, FC does not guarantee an identical interpretation anyway (see section 5.3.1), so this remains unclear. It appears problematic that elements generated by IM are no longer automatically copies; the very fact that this question can now be reasonably posed suggests a problematic shift in the framework.

An additional issue concerning FC is that it cannot guarantee copy relations at C-I, since it is subject to the PIC. Consider (84). First FC applies between *cat*₂ and *cat*₁. This is followed by Transfer of the phase head complement (PHC) indicated in grey. FC then applies between *cat*₃ and *cat*₂.

⁶⁵ *John*₁ might move to Spec CP as a topical element, which would solve the accessibility problem for FC; however, this causes issues for MY (see section 5.6.3).

(84) {... cat₃ ... {PHASE cat₂ ... {PHC ... cat₁ ...}}

Although each individual FC operation is unproblematic, *cat*₁ is transferred before the other two copies. Unless FC endows copies with a ‘copy’ feature, which is undesirable (see section 6.5.2), the copy relation between all inscriptions of *cat* must presumably be built back together in some way at C-I.⁶⁶ This suggests that FC is not particularly helpful for the semantics. Instead, it is once again an operation which allegedly provides useful information for C-I, but, in reality, serves only to support the syntax.

This issue highlights the ambiguous status of FC as a derivational operation. FC is subject to the PIC and is Markovian, which are clearly derivational properties. However, the notion of a ‘copy’ is primarily a representational one, concerning interpretation. This exemplifies the unclear status of certain elements in Chomsky’s work in relation to the syntax. I will assume that FC is a derivational operation, as this is what Chomsky intends.

5.2.2 FC-copies versus Preservation-copies

This section provides a more detailed examination of Preservation versus FC. I argue that Preservation and FC refer to different types of copies.

Chomsky et al. (2023) argue that the interpretive component can distinguish IM-copies from EM-copies using Preservation and/or thematic information (see chapter 4). As seen in the following quote, this is similar to the copy/repetition distinction in Chomsky (2019b) (repeated in (85)) but applied after FC.

⁶⁶ Relatedly, it remains unclear how and in what format FC provides the information that inscriptions are copies.

the lower copy functions as 'PRO' if its c-commanding identical element occupies a theta position; otherwise, it is 'trace.' When they function as trace or PRO is thus deducible. [...] By looking over the steps of the derivation, which Preservation by definition has access to, Preservation can detect the identity relation between [a copy and its trace] (Chomsky et al., 2023:44)

- (85) elements in theta-positions must be repetitions (i.e., EM'ed there), and those in non-theta-positions must be copies (i.e., IM'ed there). (Chomsky, 2019b:44)

Thus, a more fundamental 'copy/repetition' distinction is being made, after the application of FC, between 'true' IM-copies and EM-copies, which derives the Control/Raising distinction. This indicates that the copy relation created by FC is less fundamental than the one assured by Preservation. Control structures thus contain FC-copies but not Preservation-copies. Preservation-copies appear truly identical, unlike FC-copies; indeed, the copies in a Control structure bear different theta-roles. In a system containing the principle of Preservation, which emphasises how elements are to remain identical throughout the derivation, it is unclear why a copy relation might be assigned to two inscriptions that are not quite identical. The role of FC also needs to be clarified: what is it sensitive to, and how identical must FC-copies be?

5.2.3 FC as an externalisation operation?

Thus, it appears that Preservation creates 'true' (IM-)copies in the interpretive component (allegedly yielding the Control/Raising distinction; however, see section 4.4), whereas FC creates more 'surface-level' copies, primarily to derive the right result in the phonological component (the lower copy not being pronounced). The existence of both FC and Preservation appears redundant as they both form copies;

might it be possible to separate them into copies for interpretation (Preservation) and copies for externalisation (FC)? Although separating these two aspects of copy formation is questionable, it would at least provide a clear division of roles between Preservation and FC. This seems to fit with Chomsky et al.'s (2023) and Kitahara's (2023) argument that (86) is grammatical but unacceptable in externalisation (see sections 4.4.3 and 5.4).

(86) *John tried John to win (Kitahara, 2023:217)

This approach has obvious problems, however. FC clearly has semantic effects; forming a copy relation between two elements by definition affects their interpretation. Furthermore, Chomsky intends FC to be relevant to interpretation:

It is at this point that FC applies since the copy relation is relevant to the semantics; for example, FC contributes crucial information for the semantic interpretation of inscriptions. It is an operation that applies to a syntactic object, and which provides 'instructions' to the interfaces relevant to interpretation just at the point that the object is entering semantic interpretation; FC provides the information 'these two identical inscriptions are copies' and the semantics thus knows to interpret them in the same way. [...] [T]he information provided by an application of FC is critically important to CI (interpret structurally identical inscriptions X, Y in exactly the same way). (Chomsky et al., 2023:25-26)

FC, then, cannot be restricted to externalisation. However, its role in interpretation is unclear. How can FC ensure that inscriptions are interpreted 'in exactly the same way' if EM-copies can bear distinct theta-roles, and what purpose would an operation guaranteeing this serve? Even worse, Chomsky is inconsistent about whether the

application of FC affects theta-roles. Indeed, he uses theta-theory to constrain the output of FC. As seen in section 4.3, (87) is barred by the Univocality of theta-theory.⁶⁷

(87) *John saw ~~John~~ (= (56))

Thus, the aspect in which EM-copies can allegedly differ, namely theta-roles, is precisely what the application of FC affects. The relationship between FC and theta-roles remains unclear.

5.2.4 Theta-linking revisited

Thus, FC affects theta-roles, although many questions remain. Rugna and Manzini take this notion further, arguing that FC yields theta-linking.

Rugna & Manzini (2023) and Manzini & Rugna (2024) adapt FC-theory to new structures, namely wh-dependencies. The core theoretical claims remain consistent across both papers, with Rugna & Manzini (2023) focussing on Romance and Manzini & Rugna (2024) on Germanic. This section illustrates the argument for Germanic.

Manzini & Rugna (2024) present an FC analysis of wh-dependencies in varieties of German and Dutch. They focus on wh-copying (88), and wh-scope marking (89).⁶⁸

⁶⁷ The ungrammaticality of (87) might be argued to arise from the Theta-theory/Duality₂ violation of movement of *John*. However, it appears that (87) also cannot be derived from an M-gap analysis, with the two inscriptions EM'ed in accordance with Theta-theory/Duality₂, followed by FC. The fact that this is disallowed suggests that FC does affect theta-interpretation rather than only yielding surface deletion. How this is to be reconciled with the absence of theta-related effects in Control derivations remains unclear.

⁶⁸ The fact that the lower copies in these examples are overt is returned to in section 5.4.

(88) **Wen** glaubst du **wen** Irina liebt?

who believe you who Irina loves

‘Who do you believe that Irina loves?’

(89) **Was** glaubst du **wen** Irina liebt?

what believe you who Irina loves

‘Who do you believe that Irina loves?’ (Manzini & Rugna, 2024:2)

Manzini and Rugna argue that wh-copying and wh-scope marking involve EM of wh-elements in the scope position, rather than IM as is standardly assumed. EM analyses of phenomena traditionally ascribed to IM are becoming increasingly common; for instance, Woods & Roeper (2020) argue that child language questions such as (90), containing two auxiliaries, are best analysed as involving two instances of EM.

(90) Can he can go? (Woods & Roeper, 2020:835)

Manzini and Rugna argue that the higher wh-element is related to the lower wh-element by FC, similarly to Chomsky’s M-gap analysis of Control structures. Their work uses definition (91) of Duality₂, which allows EM to non-theta-positions (see section 3.4).⁶⁹

(91) For A-positions, EM and EM alone fills a θ -position. (Chomsky, 2021:30)
(= (33))

The higher inscription is EM’ed in a non-theta-position yet must receive a theta-role. Manzini and Rugna argue that ‘by virtue of FC [the higher inscription] can be ‘ θ -linked’ [...] despite being merged in an A’-position.’ (Manzini & Rugna, 2024:171). In

⁶⁹ Manzini and Rugna’s analysis violates Duality₁ as it involves EM yielding discourse meaning.

other words, FC yields not only a copy interpretation, but also theta-linking.^{70 71} This is not entirely inconsistent with Chomsky's work; indeed, section 5.1.2 argued that the definition of theta-linking, given its reference to copies, may interact with FC, and that this would cause problems for Control structures. It appears that Chomsky seeks to avoid this consequence by referring to Preservation-copies in his version of theta-linking. In contrast, Manzini and Rugna refer to FC-copies, leveraging the consequences to establish the correct theta-relations in wh-dependencies. Manzini and Rugna's work thus highlights the inconsistencies of Chomsky's stance on FC and theta-roles.⁷²

5.3 Copy identity

5.3.1 'Identity' for FC

The previous sections have highlighted the differences between FC-copies and Preservation-copies. I have argued that FC-copies are less 'fundamental', less 'identical', than Preservation-copies. This has been discussed in the context of theta-roles. The following section explores this further, remaining within Manzini and Rugna's work. For Manzini and Rugna, the requirement on FC is not strict identity, but rather featural non-distinctness, as defined in (92).

⁷⁰ For Chomsky, theta-linking is an A-phenomenon, whereas for Manzini and Rugna it also applies in the A'-system.

⁷¹ If the main purpose of Duality₂ is to ensure that arguments are EM'ed in theta-positions to receive theta-roles, theta-linking seems to undermine this principle. With this version of theta-linking (whereby 'copy' refers to the output of FC), it is possible for an argument EM'ed in a non-theta-position to still receive a theta-role, assuming a copy in a theta-position is available for FC to apply. This incorrectly allows for structures such as (1)

(1) *John expects to seem that Bill left (Epstein et al., 2015:65)

If Duality₂ can be bypassed in this way, it loses its predictive power.

⁷² Kitahara's (2023) work also emphasises these issues. In discussing Chomsky (2021), he argues that since FC results in copies being interpreted identically, FC yields theta-linking. As I have discussed, this is clearly paradoxical as this is not what is observed in Control structures.

(92) A and B are non-distinct if the intersection of their FFs is not empty and the FFs they share have non-contradictory values. (Manzini & Rugna, 2024:6)

This allows for the derivation of partial wh-copying (93)-(94) and wh-scope marking (95), with FC applying between the bolded elements on the basis of featural non-distinctness of their φ - and wh-/F-features.

(93) **Wen** glaubst du **den** Irina liebt?
who believe you this Irina loves
'Who do you believe that Irina loves?' (Manzini & Rugna, 2024:14)

(94) **Wie** denk je **welke jongen** ik gezien heb?
who think you which youngsters I seen have
'Which young people do you think I saw?' (Manzini & Rugna, 2024:15)

(95) **Was** glaubst du **wen** Irina liebt? (= (89))
what believe you who Irina loves
'Who do you believe that Irina loves?' (Manzini & Rugna, 2024:20)

The notion of non-identical copies was already problematic in Chomsky's work, but the issue is more significant here. This appears to be another case of a semantic notion, i.e., the establishment of a copy relation, being used primarily to benefit the syntax, a recurring theme within this work, as discussed in the introduction. Featural non-distinctness appears similar to the notion of feature sharing in labelling theory. Section 6.4.2 argues that feature sharing is a problematic notion; however, it at least requires an Agree relation, providing some predictive value. Featural non-distinctness as phrased in (92), in contrast, is too general and not sufficiently predictive.

Manzini and Rugna use the morphophonological differences between the copies in (93)–(95) as evidence for these structures not being generable by IM. This implies that identity is still required for IM-copies, unlike for EM-copies. This ties back to the notion that IM still automatically produces Preservation-copies. Due to the Markovian nature of derivations, the system cannot distinguish IM-copies from EM'ed inscriptions, so we cannot state that identity is required for FC of IM-copies, unlike for EM-copies. Instead, it appears that this is derived from the fact that IM will never yield non-identical inscriptions by Preservation. Once again, we see the interaction of these two types of 'copy' without explicit mention.

5.3.2 'Identity' for Preservation

The question of identity for FC-copies remains problematic. Preservation-copies, however, should be less of a concern, as they refer to copies in the traditional sense, but the notion of 'identity' also requires clarification here.

Firstly, application of FC would violate a very strict version of Preservation: the lower copy has been linked to the higher one, and its interpretation has thus been altered during the derivation. This version of Preservation is too restrictive.

The bigger issue concerns discourse. Elements acquire additional meaning in the positions through which they move. For instance, movement of a DP to Spec-TP gives it the interpretation 'subject of predication', and movement to the left periphery might endow it with focus, topic or interrogative meanings. Unless we adopt Aboh's (2010) approach from section 2.3, where such features are present on LIs from the numeration, this would violate Preservation. It seems instead that Preservation is concerned with the intrinsic meaning of elements; acquiring additional relational interpretations, such as discourse-related meanings, must be acceptable.

5.4 Copy visibility

One remaining issue concerning the definition of FC-copies is whether FC makes copies silent at externalisation. Indeed, in Manzini & Rugna's (2024) examples discussed above, the lower copy is overt, suggesting that FC does not necessarily guarantee copy deletion at externalisation.

I argue that Manzini and Rugna's notion of an FC-copy is different from that of Chomsky. Indeed, as discussed, Manzini and Rugna allow FC to apply between non-identical copies, and FC does not make copies silent, this is done by a separate rule. This suggests that FC is being used to derive coindexation rather than copies in the standard sense – coindexation can no longer be derived using indices, as this violates the Inclusiveness Condition (see section 4.1.1). In contrast, Chomsky treats FC-copies as identical (ignoring the issues regarding theta-roles) and uses FC to render them silent: 'SM uses the 'copy relation' information supplied by FC as an instruction to not pronounce the lower copy' (Chomsky et al., 2023:26). Chomsky et al. also distinguish coindexation from FC: 'it's possible for repetitions to be interpreted as co-referential. Thus, in *He thinks he's smart*, the identical inscriptions of *he* are repetitions, but can still have the same referent' (Chomsky et al., 2023:24). How coindexation can be derived in a theory without indices is a problem beyond the scope of this work, but Manzini and Rugna's concept of FC-copies clearly differs from that of Chomsky.

In this section, I argue that FC must make copies silent, otherwise a clear problem arises when attempting to analyse the data. Consequently, how to incorporate Rugna and Manzini's work becomes unclear.

To illustrate this, I provide three alternative explanations for the ungrammaticality of (96).

(96) *John tried John to win (= (86)/(73))

The obvious explanation for (96) is to state that FC has not applied, which leads to various issues, including Case and coindexation (and, in my approach, labelling – see section 6.3). However, Kitahara (2023) and Chomsky et al. (2023) argue that (96) is grammatical but unacceptable in externalisation. In other words, FC has not applied, but this is solely an issue for externalisation. This is problematic, as Chomsky considers FC to be relevant not only at externalisation, as I have discussed.

Manzini & Rugna's (2024) work introduces the possibility of an alternative analysis. Since FC for them does not have to involve deletion at externalisation, it becomes possible to analyse (96) as a structure in which FC has applied, but the copy has remained overt. As for Kitahara and Chomsky, the problem is at externalisation.

This notion of sentences which are grammatical but problematic at externalisation is problematic. In particular, this final approach poses significant problems for the evaluation of the grammaticality of sentences. The only possible criterion to determine whether FC has applied is the presence or absence of an overt copy. For instance, in examples (97)-(98) it is assumed that FC must apply to avoid theta-related issues. However, if FC does not guarantee deletion, it could be possible that FC has applied, but not rendered the copies silent – the ungrammaticality pertaining solely to externalisation. If FC does not guarantee copy deletion, it is unclear why (97)-(98) are disallowed.

(97) *John seems John to be happy

(98) *John was seen John

An approach in which FC does not guarantee silent copies lacks predictive power. There is no clear way to determine whether FC has applied in (96), and thus if the sentence is ungrammatical or solely a problem at externalisation. This approach, despite purportedly allowing FC-copies to surface as overt copies, leaves unexplained why this sometimes causes externalisation problems, but is at other times allowed, as in Manzini and Rugna's examples.

For these reasons, I will assume that FC-copies must be silent. How this fits with Manzini and Rugna's work, and with other cases of overt copies, is a question which remains open for future research.

5.5 Interim summary

The previous sections have introduced the notion of FC-copies and Preservation-copies and discussed issues that arise in MERGE with the introduction of FC. It has become obvious that the definition of a 'copy' has become increasingly unclear.

Figure 1 summarises the various interpretations the word 'copy' has been given in MERGE and how these different definitions interact. The main division is into FC-copies and Preservation-copies, with overlap for IM: indeed, IM without FC is impossible. This is no longer an automatic consequence of the theory, an issue discussed in section 5.2.1. 'Coindexation' is used to describe copies in the sense of Manzini & Rugna (2024), i.e., elements related by FC, despite having to be neither identical nor covert.

It is worth noting that in MERGE, as seen in sections 1.1.1 and 1.1.3, copies are formed automatically by Merge.⁷³ This is derived from Preservation (since creating $\{[a, b], a, b\}$ from $[a, b]$ arguably involves minimal alternation of the workspace, although I remain unconvinced by this, see footnote 9). Preservation-copies only persist in cases of IM due to MY, however.⁷⁴

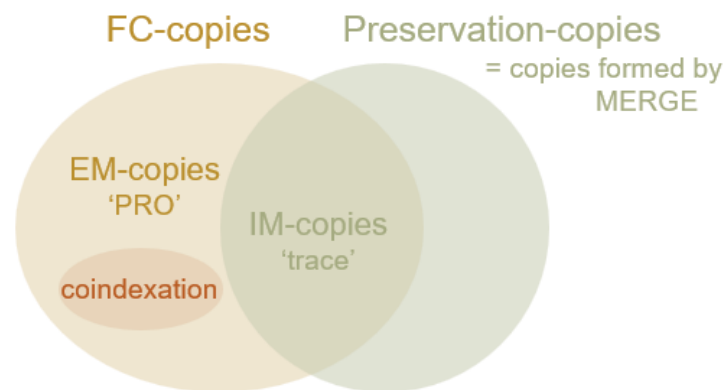


Figure 1. *Different interpretations of ‘copy’ in MERGE.*

Thus, the definition of ‘copy’ has become undesirably complex. Furthermore, the use of the same term to refer to at least five different concepts, without explicit clarification, renders work written within the MERGE framework challenging to read.

A further issue concerns redundancy. Indeed, if Preservation keeps track of IM-copy relations, it is unclear why the FC operation is necessary at all. Chomsky insists that the system is Markovian, so there is no notion of a copy relation until FC applies. However, the introduction of Preservation suggests the exact opposite. Indeed, despite Chomsky’s insistence that Preservation is irrelevant to the syntax, it has been used to constrain Merge in the context of Duality₂ and to explain the syntactic

⁷³ This means that MERGE involves more than just combining elements, which seems undesirable in a theory seeking simplicity. This remains an open question for future research. This also means that technically, everything produced by Merge is a (Preservation-)copy (although the original copies have been deleted in cases of EM), yet nothing is considered a (FC-)copy until FC applies.

⁷⁴ This implies that application of MY does not violate Preservation.

differences between Control and Raising derivations. Preservation is intended to pertain solely to interpretation, thus justifying the need for FC-copies; the fact that Chomsky is inconsistent about its relevance to the syntax is concerning.

A concession must be made, however. As MERGE is a relatively new framework, concepts have been introduced incrementally without full formalisation or consideration of their interactions. For instance, Chomsky (2019b) introduced a new copy/repetition distinction, which is invalidated by the introduction of FC. The introduction of Preservation in turn somewhat undermines FC, as discussed above. Over time, a clearer and more coherent understanding of ‘copies’ in MERGE might arise as the relationships between these concepts are more thoroughly explored. The current proliferation of definitions of ‘copy’ stems from the gradual development and lack of formalisation of these ideas. However, in its current form, copy theory in MERGE is fraught with inconsistencies and faces numerous challenges.

5.6 Copy accessibility

The following section discusses another issue arising from the fact that copies no longer come for free in MERGE. This pertains to MY. As discussed in section 1.1.3, MY rules out extensions of Merge (e.g., Parallel Merge, Sideways Merge) as these involve creating two accessible terms while MY allows at most one. Chomsky argues that IM does not violate MY, despite it creating an additional copy. Indeed, MS and/or the PIC render the original copy inaccessible to Merge.

This section criticises this idea. First, I discuss whether MY refers to Preservation-copies or FC-copies. Indeed, Chomsky intends ‘copy’ to refer specifically to FC-copies; however, in the context of MY, it appears that ‘copy’ refers to Preservation-

copies. The following section argues that MS is unsuccessful at preventing IM from violating MY, although the PIC may be sufficient.

5.6.1 MY refers to Preservation-copies

IM obeys MY because the lower copy is inaccessible by MS. In the structure $\{a \{a, b\}\}$, only the first inscription of *a* is visible. MY was introduced in Chomsky (2019a), i.e., in the framework in which copies still arose automatically. Now, given the existence of FC, determining that the two inscriptions of *a* are copies requires either FC to have applied, or reference to be made to Preservation. This section argues that MY refers to Preservation-copies, not FC-copies.

If MY refers to Preservation-copies, the copy relation comes for free without the need for a rule to apply before determining whether the derivation conforms to MY.

However, this is problematic because Preservation occurs at interpretation. At the stage at which MS is determining whether the WS' satisfies MY, the derivation is Markovian. As already discussed, having Preservation apply in the syntax is deeply undesirable.

The advantage of using FC is that it operates in the syntax, not at interpretation. However, FC would have to apply immediately, rather than at the phase level as Chomsky intends. Having operations apply to the WS' before MY seems undesirable. Intuitively, it also appears that we are dealing here with copies in the more 'fundamental' sense of Preservation-copies. This becomes clearer when considering the different implications for accessibility posited by the two options. Consider (99)-(100):

(99) $WS = [\{a, b\} \dots]$ – IM of a

Merge($a, \{a, b\}, WS$)

$WS' = [\{a \{a, b\}\} \dots]$

(100) $WS = [a, \{a, b\} \dots]$ – EM of a

Merge($a, \{a, b\}, WS$)

$WS' = [\{a \{a, b\}\} \dots]$ (i.e., $[\{a \{a, b\}\}, a \dots]$ with the outer a deleted by MY)

I first illustrate the consequences with FC-copies.

In (99), the innermost a is rendered inaccessible by FC. In (100), the outer a is deleted as noted in the parentheses. If FC applies here,⁷⁵ a copy relation is established between the two a 's in $\{a \{a, b\}\}$.⁷⁶ This renders the innermost a inaccessible. This predicts a different cardinality of the workspace for structures with M-gaps compared to those with repetitions or other EM'ed elements. While the implications of this are beyond the scope of this work, it seems intuitively problematic. If we instead refer to Preservation-copies, the innermost a in (100) remains accessible, as it is an independently merged element, which appears more plausible. It also seems more coherent with Chomsky's phrasing, since he states that 'Merge automatically produces copies. Because of [MY], they only persist in applications of IM.' (Chomsky, 2024:24). 'Copies' here clearly refers to Preservation-copies.

Thus, it appears that MY is best understood as referring to Preservation-copies.

However, this presents a problem, as Preservation applies at interpretation. Neither

⁷⁵ For consistency with the IM case, FC would apply before MY, i.e., before deletion of the outer a .

⁷⁶ No FC relation is created with the outermost copy, since it does not c-command the structure.

approach is fully satisfactory. It is, therefore, not surprising that Chomsky's work incorporating FC does not offer a detailed explanation of how IM conforms to MY.

5.6.2 MS does not render copies invisible

A further problem exists with Chomsky's approach to IM and MY as it is unclear why MS would render the lower copy inaccessible. Indeed, in a structure such as $\{a \{a, b\}\}$, MS locates the outermost a , then proceeds to locate b . Since it reaches the level of embedding of b anyway, it is unclear why it would not also locate the lower a .

Mizuguchi (2023:52) also argues that MS is unavailable at the stage where MY applies. Indeed, MY applies to the output of Merge, whereas MS 'is possible only when operations actually apply in the next step or apply to the input of MERGE, not when the WS is mapped to the WS' by MERGE.' Indeed, the notion of operations such as MS (or FC as discussed in 5.6.1) applying before MY appears problematic.

If MS cannot enable the structure to conform to MY, it is worth examining whether this can be done entirely by the PIC, as this is the other solution Chomsky proposes. Mizuguchi (2023) argues for precisely such an approach, although for different reasons to mine. MY can be guaranteed by the PIC if we assume that Transfer applies freely, not only at the phase level. This is not entirely uncontroversial, with, as Mizuguchi mentions, Chomsky having stated similar ideas: 'One plausible element of optimal design is that there are no constraints on applications of operations.

Accordingly, SPELL-OUT can apply anywhere' (Chomsky, 1998:121-122). Thus, Transfer would presumably apply after each instance of IM to ensure that the derivation conforms to MY. The implications of free Transfer for accessibility are likely problematic, though a discussion of this is beyond the scope of this work. It is also unclear why free Transfer could not be applied in a similar way to permit extensions

of Merge. I return to the concept of free Transfer in section 6.5.1 in the context of labelling theory.

5.6.3 Remnant movement

Thus, the MS-based approach to MY is fundamentally flawed. Nevertheless, this section discusses how it addresses remnant movement. As alluded to in section 5.2.1, remnant movement is a potential concern for FC-theory. Epstein et al. (2021) argue that the grammaticality or ungrammaticality of structures involving remnant movement can be explained by whether they violate MY. At first glance, all remnant movement appears problematic due to the presence of two accessible copies. However, grammatical cases are possible, they argue, as the PIC can render one of the copies inaccessible. This is illustrated in (101), where there is only one accessible copy, the greyed-out structure rendered inaccessible by the PIC.⁷⁷

(101) I wonder how likely to win John is (=83)

{CP {AP how likely {~~John~~₃ to win}} {C {IP John₁ {is {AP how likely {~~John~~₂ to win}}}}}}}

(Saito, 2022:172)

However, it is unclear to me how two copies could ever be accessible, given the assumption that MS stops once it has located the first copy. This would imply that multiple copies are accessible only when they are both located by MS, i.e., when they are merged at the same level of embedding. For illustration, Epstein et al. argue that (102) violates MY as two copies are accessible. (103) illustrates the derivation up to the merger of *wonder*.

(102) *Which picture of does John wonder whom Mary bought?

⁷⁷ If *John* moves to Spec CP to receive topic meaning, this analysis does not hold for this example.

(103) {wonder {{which {picture of **whom₃}}** {**whom₂** C Mary bought {~~which {picture~~
of **whom₁}}**}} (Epstein et al., 2021:132)

It is not evident how two copies are to be accessible in this example, given that *whom₂* is located first, and MS is assumed to stop after finding the first copy.

Sugimoto (2022:59) claims that the lower copy in such structures is accessible because there is no c-command relation between the copies, however, it is unclear why this would affect the halting of MS.

A detailed discussion of remnant movement is impossible for space reasons.

However, I have demonstrated that incorporating remnant movement into MERGE is not straightforward. Remnant movement is a problem for FC-theory as discussed in section 5.2.1, and the relationship between remnant movement and the accessibility of copies remains unclear. Further research is required to integrate remnant movement into the MERGE framework.

5.7 Markovian derivations

The final section of this chapter addresses the implications of the claim in MERGE that derivations are Markovian, as introduced in section 1.1.4. MERGE, and other computational operations applying in the syntax, are strictly Markovian, with no memory of past stages of the derivation. This is stricter than past work, which allowed for phase-level memory (see section 4.1.3).

Further research is needed to determine whether all operations in the syntax are genuinely Markovian, a topic beyond the scope of this work. This issue is closely tied to the ambiguous status of certain phenomena in relation to the syntax. I have already noted various cases of such ambiguity, including FC. Given that derivations are Markovian while interpretation (which examines the representation) is not, this

uncertainty is now particularly problematic, as the status of a rule or phenomenon in relation to the syntax affects what it can access. Of particular concern is labelling. As I discuss in section 6.2, the status of labelling in relation to the syntax is particularly unclear. This issue is also highlighted in Chomsky's discussion of the differences between Control and Raising derivations (see section 4.4), where he states that 'INT cannot distinguish them, but other operations can, and do' (Chomsky, 2021:22). 'INT' here refers to 'an expository device to lay out the options for the next step of derivation' (Chomsky, 2022:361), i.e., to the syntax (see footnote 51). If 'operations' here refers to syntactic operations, they should also be unable to distinguish the two structures, so, as discussed, it appears that Chomsky considers the Control/Raising distinction interpretive. However, the use of the term 'operation' here obscures this somewhat, as this is often (but not necessarily) used to refer to syntactic operations.

With derivations being strictly Markovian, MERGE introduces a significant difference between derivational and representational perspectives on the same structure. As Chomsky (2021:20) notes, the workspace 'does not contain items that were generated earlier' and 'the history is not preserved in the current state'. In earlier work, derivational and representational approaches were often considered metaphors which described the same underlying processes. However, in MERGE, the representation appears to be fundamentally distinct from the derivation. The theoretical and empirical implications of this require further research.

5.7.1 Bošković (2024) on Markovian derivations

This section discusses Bošković (2024), as it appears at first glance to challenge the notion of derivations being Markovian. If this is correct, and derivations are not Markovian, this could pose significant issues for MERGE, invalidating phenomena

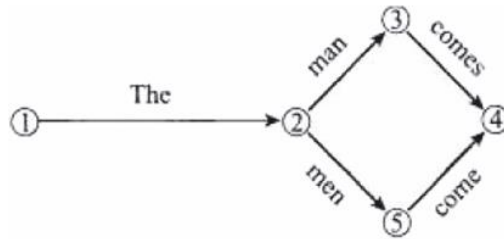
such as M-gaps.⁷⁸ Bošković cites Chomsky (2021) but does not discuss in any depth the implications of his approach on MERGE. I argue that Bošković's and Chomsky's uses of 'Markovian' refer to different concepts.

Bošković argues that many elements of the syntax are contextual, including the EPP and the A/A' distinction (see section 2.4.4). These cannot be defined based solely on Merge creating the relevant positions (for instance, A-positions are traditionally defined as positions merged to a Case or theta-assigning head).⁷⁹ Contextuality does not function in a framework with Markovian derivations, as it is necessary for the system to remember the history of the derivation for contextuality to be meaningful. In other words, the syntax cannot be Markovian. Another reason Bošković gives for this is the existence of non-adjacent relations in language.

Bošković argues that this aligns with Chomsky's (1957) claim that language cannot be captured by a finite-state device. Finite-state devices are defined by the following features: an initial state, a finite number of final states, a specification of transitions between states, and a specification of the symbol printed when a transition occurs (Bošković, 2024:49). (104) illustrates a finite-state device grammar generating the phrases *the man comes* and *the men come*.

⁷⁸ In fact, this is not necessarily the case. Indeed, Collins & Groat (2018) discussed the issues with phase-level memory approaches to EM'ed vs IM'ed inscriptions back when memory was available at the phase level, arguing that it is unclear how such information is to be stored (see section 4.1.3).

⁷⁹ Bošković likely considers theta-positions to also be contextual, which could have interesting implications on the definition of Duality₂, particularly on the idea of A'-'theta'-positions (see section 3.4.2).



(104)

(Bošković, 2024:49)

However, it becomes immediately clear that Bošković's definition of 'Markovian' is different from that used in MERGE. Indeed, MERGE is not a finite-state device. At any given stage, finite-state devices cannot access or take into account anything that has been generated previously. MERGE, however, can see elements present in the workspace, even if does not have access to the history of how they were derived (e.g., if they were IM'ed or EM'ed). This is illustrated in (105). The outer / is deleted by MY (indicated in grey), rendering the derivation Markovian. But the workspace still contains an inscription of /.

(105) WS = [{/}, {see {see {the cat}}}]

Merge({/}, {see {see {the cat}}}, WS)

WS' = [{/}, {/ {see {see {the cat}}}}]

As I have already illustrated in several instances throughout this thesis, this example further highlights the terminological confusion which can arise when concepts are not clearly defined. In MERGE, Chomsky is using 'Markovian' differently from in previous work. Consequently, although Bošković (2024) may initially appear to present a significant challenge to MERGE, it is actually criticising an entirely different notion of 'Markovian' derivations.

However, even if Chomsky were claiming that MERGE is Markovian in the sense of Bošković (2024), he might argue that Bošković's evidence does not actually pose a problem for his theory. The fact that MERGE cannot capture non-adjacent relations

is unproblematic, as MERGE is a free operation, blindly combining elements. At the interfaces, the interpretive component is non-Markovian and can filter out all impossible derivations. Furthermore, given that Chomsky has suggested that Agree might no longer pertain to the syntax (see section 6.4.2), non-adjacent agreement would not be a problem either.

Theories of entirely free Merge are not without their problems, however. In the search for simplicity and elegance, Merge has been made maximally simple, yet this simplicity appears excessive. Concepts like the A/A' distinction and the EPP are syntactic in nature,⁸⁰ and should not be relegated to the interfaces in this way. Although this issue is beyond the scope of this thesis, the question of how free MERGE should be and how many phenomena can reasonably be relegated to the interfaces is certainly worth further consideration.

5.8 Summary

This chapter has explored the theoretical implications of the introduction of an FC operation and its interactions with Preservation. IM-copies no longer come for free, but the consequences of this have not yet been fully considered. This not only results in a proliferation of different elements referred to as 'copies', which creates confusion and inconsistencies in the theory, but also impacts Chomsky's argument that IM does not violate MY. I have discussed various problems with IM and MY, including how MS fails to account for lower copy invisibility. These issues do not seem to have a ready solution.

⁸⁰ Although Chomsky is ambivalent on the status of the A/A' distinction in his current work, potentially attempting to derive it from Duality, or stating outright that it may not be theoretically valid.

FC being Markovian and Preservation non-Markovian, this discussion also ties into the question of which phenomena pertain to syntax versus interpretation, and to larger questions surrounding the Markovian nature of computation. The fact that representational and derivational perspectives on the same structure are now so different is a significant shift from previous work, and this area is yet to be explored.

6 Reconceptualising copy visibility under FormCopy

The introduction of FC has significantly altered the status of copies, with wide-ranging implications. This chapter examines the implications of FC-theory for labelling theory (Chomsky, 2013; 2015). Although labelling theory is the direct predecessor of MERGE, the interaction between FC and labelling has not yet been investigated in detail.

To begin, I briefly overview labelling theory. In Chomsky's (2013, 2015) work, the focus is on how constituents can be labelled. With Merge not involving projection, labelling is achieved through the Labelling Algorithm (LA), which is reduced to MS. At the phase level, the LA selects the most prominent LI as the label. Labelling is thus straightforward in $\{X, YP\}$ configurations. When Merge generates symmetric $\{XP, YP\}$ structures, however, the LA cannot determine a label since both LIs are equally prominent. To resolve this, one of the constituents undergoes movement, allowing the remaining constituent to provide the label. The moved constituent continues to move until it reaches a configuration where there is no longer a labelling failure, i.e., a configuration of Spec-head feature sharing. Indeed, Chomsky argues that if two elements share features and can participate in Agree, the shared features can provide a label for the structure. Chomsky has applied labelling theory to, amongst other things, vISH and successive-cyclic wh-movement.⁸¹

For labelling theory to function, when a constituent moves, its lower copy must become invisible for labelling. I argue that this is not only stipulative but is invalidated

⁸¹ Labelling theory is also relevant in the case of $\{X, Y\}$ constructions, which is beyond the scope of this work. See Dadan (2019) and Moro & Roberts (2023) for further discussion.

by MERGE's assumptions on copies. I suggest that instead, FC renders copies invisible for labelling.

Other issues with labelling theory remain, however. These concern the overarching theme of whether labels pertain to syntax or interpretation. I argue that labelling theory claims that labels pertain to interpretation, yet requires them to be relevant to the syntax. Indeed, the motivation given for labels is semantic, yet the information they provide is useful primarily for the syntax. I discuss this in the context of lower copy invisibility and feature sharing. I provide some tentative solutions to these problems, including adapting Bode's (2020) approach; however, none of these are without problems. I argue that this illustrates a fundamental problem with labelling theory.

The final section illustrates the implications of integrating labelling theory with FC-based approaches, arguing that, although it is possible to unify both theories, the individual problems concerning FC and labelling remain. I present illustrations of structures which have been analysed in FC-theory, adapted to my framework. My approach makes similar empirical predictions to labelling theory, though there are significant outstanding issues.

6.1 Lower copy invisibility

The notion of lower copy invisibility, essential for labelling theory, was first mentioned indirectly in Chomsky (2008:150), where he argues that 'an A-chain becomes invisible to further computation when its uninterpretable features are valued.' 'Computation' here includes Agree, and presumably labelling. This stipulation requires the outdated notion of a chain (see section 4.1.2) and originates from an earlier framework in which features were a lot more important to the syntax.

In labelling theory, lower copy invisibility follows from copies being part of a discontinuous element.

The intuitive idea is that the lower XP copy is invisible to LA, since it is part of a discontinuous element, so therefore [the symmetric structure] β will receive the label of YP. The phenomenon is similar to intervention effects, where the head of a chain (more appropriately, the whole chain, the discontinuous element) induces an intervention effect, but not the lower copy. (Chomsky, 2013:44)

However, lower copy invisibility is never properly justified beyond this 'intuitive idea'. It remains a stipulation, essential for labelling theory to work.

Epstein et al. (2021) attempt to derive lower copy invisibility for labelling by arguing that it is derived from the same source as lower copy inaccessibility in the context of MY: as seen in section 5.6, MS is argued to find the first copy and then stop searching. I argued in section 5.6.2 that it is unclear why MS would not find the lower copy, given that it needs to search that level of embedding anyway. I illustrate this in the context of labelling with (106).

(106) $\{\beta \text{ DP TP } \{T \{\alpha \text{ DP vP}\}\}\}$

Merger of DP and vP creates an unlabellable symmetric structure α . After movement of DP, the lower copy is allegedly now inaccessible though MS, and thus invisible to the LA, so α is labelled vP. However, it is unclear why MS cannot see the lower DP, given that it scans that level of embedding anyway to locate vP. Lower copy invisibility must still be stipulated.

It is also worth noting that Epstein et al. (2021), in using the same justification for copy invisibility for MY and labelling, unintentionally commit themselves to an

approach in which labels are relevant to the syntax.⁸² Indeed, MY is clearly syntactic rather than interpretive. Whether labels pertain to the syntax or the interfaces is extremely ambiguous in current work, as discussed in the following section.

6.2 Is labelling syntactic?

With the introduction of FC, IM-copies no longer arise automatically in the syntax. Preservation still keeps track of IM-copy relations, but this happens at interpretation. Thus, the question of where labelling applies becomes particularly relevant (see section 5.7). Indeed, if labelling is non-Markovian, it can refer to Preservation-copies, and the idea of movement deriving lower copy invisibility might still hold. However, if labelling pertains to syntax, it is Markovian and can only refer to copies in the sense of FC-copies. I argue that labelling, despite Chomsky's claims to the contrary, remains closely connected to the syntax. Furthermore, in Chomsky's new work, 'copy' tends to refer to FC-copies. If labelling involves FC-copies, movement can no longer derive copy invisibility, the consequences of which are discussed in section 6.3.

Chomsky seeks to simplify the syntax by removing as many components as possible from Merge, including labels. This was originally proposed by Collins (2002) and Seely (2006) and was adopted by Chomsky, but inconsistently. Chomsky's (2013, 2015) approach involves a tension between the wish for labels to be absent from the syntax and their role in triggering movement.

The existence of labels is justified because they are necessary for the interfaces. They are required at SM as they are involved in, for instance, prosodic rules, and at

⁸² It thus also inherits the problem with the MY approach to IM, namely that it refers to Preservation-copies despite these not being available at this point.

C-I, where they are needed, for example, to identify predicates and arguments (Bode, 2020). Chomsky (2008) illustrates the necessity of labels with (107), which, if labelled C, is an interrogative clause (e.g., *I wonder what you wrote*), and if labelled N/D, is a free relative (e.g., *I read what you wrote*).

(107) $\{\alpha \text{ what } \{C \{ \text{you wrote} \} \} \}$ (Chomsky, 2008:145)

Thus, Chomsky argues that labels pertain to the interfaces and are not syntactic in nature. However, this claim does not stop him from asserting that syntactic operations are driven by labels. This is clear in the following quote from Chomsky (2004a:109, emphasis mine).

In the worst case, the label is determined by an explicit rule for each choice of α , β . A preferable result is that the label is predictable by a general rule. A still more attractive outcome is that *L requires no labels at all*. [...] *If operations are “driven” by labels*, as we expect, then there can be no general Spec-head relation at all

Labels are explicitly talked about as driving computation until Chomsky (2013), i.e., labelling theory, where they are no longer a prerequisite for entering into computation. Indeed, this would pose a problem for cases such as (108), where an element is EM'ed to a structure which has not yet been labelled (Chomsky, 2013:43).

(108) $\{T \{ \alpha \text{ DP vP} \} \}$

This is a further step towards moving labels out of the syntax, with the aim of simplifying the first factor. Labelling theory also aims to simplify the syntax by suggesting that movement is no longer triggered by features but rather is required for labelling. Movement is claimed to apply freely, with incorrect derivations filtered out

at C-I. For instance, if DP in (108) does not move to Spec-TP, α cannot be labelled and the derivation does not converge at C-I.

However, it is not entirely clear that this involves moving labels out of the syntax.

Labels still appear to function as movement triggers in practice. Although movement is technically free, labels are key to yielding the correct derivations. Labels remain very relevant to the syntax; in practice, the difference between this approach and one that explicitly states that labels are syntactic and trigger movement is unclear.⁸³ In labelling theory, there is thus a contradiction between the claim that labels pertain to the interfaces, and their role in yielding the correct derivations of structures.

This tension is highlighted in Bode's (2020:70) summary of how recent minimalist approaches view labels. In some, the syntax is triggered, maybe even crash-proof, and labels are syntactic, driving the computation (e.g., Zeijlstra, 2020; Horstein & Nunes, 2008). In others, labels pertain to interpretation and the syntax is free. I argue that labelling theory, despite claiming to be the latter, is an in-between approach. This becomes clear when considering the work of Collins (2002) and Seely (2006), who originally proposed that labels be removed from the syntax, and their later work (Collins & Seely, 2020).

Collins (2002) argues that incorporating labelling into Merge renders its definition much more complex than desired, and discusses how various phenomena typically explained using labels can be accounted for without them. Seely (2006) adapts Collins' work, considering the elimination of labels from a derivational perspective,

⁸³ There is a difference, however, between this and approaches which require immediate labelling in the syntax as opposed to labelling at the phase level (see Epstein et al. (2015) for discussion).

arguing that labels as defined in Chomsky (1995b) are not syntactic objects, and are thus inaccessible to syntactic operations.

Collins & Seely (2020) claim that their approaches are coherent with labelling theory, arguing that labelling theory, despite referring to labels, involves an entirely label-free syntax. They claim that terms such as ‘labelling algorithm’ or ‘labels’ are used when discussing derivations, but do not have any theoretical status within the syntax.

Furthermore, ‘[m]ovement (i.e. Internal Merge) applies freely, but when it ‘must’ or ‘cannot’ apply is determined, in large part, by labeling’ (Collins & Seely, 2020:5). The use of quotation marks here clearly highlights the ambiguity of whether movement is a trigger. Collins and Seely also discuss some authors’ (e.g., Rizzi, 2015)

interpretation of labelling theory as involving syntactically represented labels and argue that this is a misunderstanding. However, the reason for such

‘misunderstandings’ is probably the ambiguity and contradictions concerning the status of labelling in the theory. Although Collins (2002) and Seely (2006) attempt to fully eliminate labels from the syntax, I argue that, contrary to Collins & Seely (2020), labelling theory still makes reference to labels in the syntax, despite claiming otherwise.

6.3 FC yields lower copy invisibility

Given that labelling still refers to the syntax, and that ‘copy’ in Chomsky’s new work tends to denote FC-copies, I assume that ‘copy’ in the context of labelling means FC-copies. This has significant consequences for labelling theory. Movement can no longer derive copy invisibility. Indeed, at the point of labelling, the system cannot tell whether an element has been EM’ed or IM’ed, given the Markovian property of derivations.

This crucial assumption of labelling theory is a significant challenge for its unification with MERGE. Both MERGE and labelling theory face serious issues; however, it is still worth examining the consequences of their unification, regardless of whether they are theoretically sound. As such, I suggest (109) as a possible solution to lower copy invisibility under MERGE.

(109) FC renders lower copies invisible for labelling.

If copies are now formed by FC rather than by movement, the natural assumption would indeed be that FC, rather than movement, derives copy invisibility: symmetric {XP, YP} structures are resolved through application of FC.⁸⁴ This has slightly different predictions than labelling theory, given that FC can form EM-copies as well as IM-copies. (110) illustrates the embedded clause of a Control derivation.⁸⁵

(110) She wants to be happy

$\{\beta \{DP \text{ she}\} \{TP \text{ to } \{\alpha \{DP \text{ she}\} \{VP \text{ be } \{VP \text{ be } \{AP \text{ happy}\}}\}}\}}\}$

This derivation includes two unlabellable structures indicated by the Greek letters. Chomsky (2013, 2015) bypasses this issue by controversially analysing PRO as a head. In FC-theory, however, it must be a phrase: what is standardly PRO is an identical inscription related to the higher one by FC. If FC renders lower copies invisible for labelling, the FC-relation created between *she* in the higher clause and *she* in (110) renders the lower copies invisible, labelling β and α TP and vP respectively. This may partially explain why FC is obligatory in Control structures, which Landau (2024) cites as an issue with FC-theory (see sections 4.4.3 and 5.4): the absence of an FC-relation in (110) would cause labelling issues.

⁸⁴ The possibility of feature sharing is discussed in section 6.4.2.

⁸⁵ See section 6.6.1 for more detailed discussion, including movement through Spec CP.

Two further points require clarification. (109) implies that FC applies freely but its output is subject to labelling. The output of FC is also subject to theta-theory (see section 6.4.3). Incorrect application of FC causes both theta- and labelling-related issues; this requires further consideration, given that theta-roles and labelling do not form a natural class.

Secondly, this approach has slightly different implications on ‘triggered’⁸⁶ operations. Indeed, labelling is no longer a movement ‘trigger’, but rather a ‘trigger’ for FC. This may be slightly less problematic than ‘triggering’ a structure-building operation; however, FC is still a syntactic operation, and one which Chomsky claims applies freely. In practice, however, this approach still requires Merge (EM or IM), given that FC requires at least two identical inscriptions. However, ‘triggering’ Merge, rather than specifically IM, may be somewhat less problematic. Indeed, labelling theory retains traces of the notion that IM must be triggered even when EM is free, contrary to my approach.

6.4 Some semantic problems with labelling theory

Thus, the general mechanics of labelling theory can be reconciled with MERGE under the assumption that FC, not movement, renders lower copies invisible for labelling. However, the structures generated by labelling theory face significant interpretive issues, linked to the central concern with labelling, namely that it is supposedly necessary for the semantics yet used primarily to yield the correct syntactic derivations. This section discusses two such issues, namely lower copy invisibility and feature sharing, and assesses whether the approach presented in (109) fares any better.

⁸⁶ I use the scare quotes since Chomsky claims that there is no trigger.

6.4.1 Lower copy invisibility

Section 6.1 argued that lower copy invisibility is inherently stipulative, but the problem is even greater. Labelling is considered necessary for an object to be interpreted at the interfaces, but if lower copies are invisible for labelling, how can they be interpreted? It must be stipulated that lower copies are invisible when it comes to labelling further structure but remain visible themselves. This requires fully committing to labels being syntactic, with a notion of visibility in the syntax (i.e., for labelling the syntactic structure) separate from visibility at C-I. This goes against Chomsky's claims, so it is unclear how lower copies can be interpreted.

Lower copies must be interpreted. This is typically illustrated with reconstruction effects. For instance, (111) is grammatical despite the surface structure seemingly violating Principle A of Binding Theory, as the lower copy of *himself* is bound by *John*, and (112) allows two readings depending on which copy is interpreted (see section 4.4.1).

(111) Which book about himself does John like ~~which book about himself~~?

(112) How many students were expected to fail the exam? (=60))

This is a significant issue in labelling theory, and my approach is even more problematic here. Indeed, (109) yields lower copy invisibility with M-gaps, which is clearly undesirable since, as discussed in chapter 5, EM'ed FC-copies are not necessarily identical (e.g., regarding theta-roles).

Takita et al. (2016) also note that copy invisibility for labelling violates the No-Tampering Condition (NTC)⁸⁷, as Chomsky (2005:13) states that '[t]he no-tampering condition also entails the so-called copy theory of movement, which leaves

⁸⁷ The NTC states that 'Merge of X and Y leaves the two SOs unchanged' (Chomsky, 2008:138).

unmodified the objects to which it applies'. Turning a copy invisible for labelling is arguably a form of modification. In MERGE, the NTC is derived from Preservation (Chomsky et al., 2023:56). Copy invisibility seems to violate Preservation, since labelling appears to be syntactic. Furthermore, given that copies are intended to be the same element, it is unclear why one copy should behave differently from another.

I discuss potential solutions to these issues in section 6.5, after considering feature sharing; however, these approaches come with their own challenges. One final comment on FC and label invisibility: If FC, applying at Transfer, renders copies invisible for labelling, it presumably must also make them invisible at externalisation. Indeed, labels are considered necessary for both interfaces. The correct labelling is derived, in some cases, through the invisibility of lower copies. Transfer then sends the structure to both interfaces. If the invisibility of a copy is responsible for yielding the correct labelling, which is relevant at externalisation, then having this copy be overt at externalisation would be inconsistent. This is a further argument for FC rendering copies silent, and an issue for approaches allowing overt lower copies (see section 5.4).

6.4.2 Feature sharing

In labelling theory, symmetric structures are labelled either by movement or by feature sharing, which occurs when the features of the two constituents undergo Agree. For instance, the constituent traditionally labelled TP, i.e., $\{\alpha \text{ DP TP}\}$, is labelled $\langle \varphi, \varphi \rangle$ by the agreeing features of DP and TP (Chomsky, 2015:9). Such labels, however, are not particularly useful for C-I, instead being relevant to the syntax, where they enable the correct derivations. This is the same problem as with lower copy invisibility. For example, Chomsky argues that in some languages (e.g., Italian), T is 'strong' enough to label the structure as TP. Aside from various issues,

including the vagueness of the notion of ‘strength’ (see section 6.5.2), this approach involves labelling as (113) TP and (114) as $\langle \varphi, \varphi \rangle$. The difference between these two structures seems to be syntactic rather than semantic, however.⁸⁸

(113) Vedo due gatti
 see.1SG two cats
 ‘I see two cats’
 {TP vedo {VP ~~vedo~~ {VP ~~vedo~~ {DP due gatti}}}}

(114) I see two cats
 { $\langle \varphi, \varphi \rangle$ {DP I} {TP T {VP {~~DP I~~} {VP see {VP see {DP two cats}}}}}}}

Feature sharing introduces other issues. Firstly, labelling by feature sharing involves Agree. In recent work, Chomsky has suggested that Agree may not be a syntactic operation (Chomsky et al., 2019:252). Discussion of this is impossible for space reasons, but Agree cannot be removed from the narrow syntax as long as feature sharing exists as a possible labelling strategy, since labelling, as argued, remains syntactic.

Feature sharing may also pose problems for Duality₂ and theta-theory more generally. Indeed, Chomsky (2013:44) suggests that labelling is necessary for theta-marking. This is presumably because labels are required to ascertain which configuration elements have been merged in, and theta-role assignment is

⁸⁸ Chomsky (2015:9) suggests that such structures do not involve *pro*. This is far beyond the scope of this discussion; I simply illustrate the structures according to his assumptions for the sake of illustration. Unless *pro* is treated as a head, similarly to the way that Chomsky accounts for PRO, the inclusion of *pro*, or an overt subject for that matter, affects the labelling in ways which Chomsky does not discuss. Indeed, it is unclear how (3) can be labelled TP, as Chomsky intends, as it is a symmetric structure.

(3) Due gatti mangiano il pesce
 Two cats eat.3PL the fish
 ‘Two cats eat the fish’
 { α {DP due gatti} {TP mangiano {VP {~~DP due gatti~~} {VP mangiano {VP mangiano {DP il pesce}}}}}}

configurational. Feature sharing may complicate the traditional understanding of a theta-position. Indeed, in (115), it is unclear whether *her* counts as being EM'ed in a patient theta-position given the $\langle \varphi, \varphi \rangle$ label.

(115) He persuades $\{\langle \varphi, \varphi \rangle \{DP \text{ her} \} \{VP \text{ persuades} \{CP \text{ to be happy} \} \} \}$

Feature sharing also highlights general issues concerning the nature of labels.

Whether labels are categorial features, feature bundles, some or all the features of an LI, or the LI itself,⁸⁹ as in bare phrase structure (Chomsky, 1995a), is ambiguous. Currently, labelling theory appears to consider labels to be categorial features unless feature sharing takes place, but further consideration is required.

6.5 Tentative solutions

Lower copy invisibility and feature sharing both present significant challenges to labelling theory. This section attempts to provide some tentative solutions, still while trying to unify labelling and FC-theory, but ultimately proves unsuccessful. I argue that this highlights the issues with labelling theory, as no clear solutions exist to address what is a fundamental flaw.

6.5.1 Adapting Bode (2020)

The first potential solution involves adapting Bode's (2020) approach to labelling. Bode eliminates both feature sharing and the notion of lower copy invisibility. Furthermore, Bode's discussion has similarities to Chomsky's work on FC and Duality₂, suggesting that adapting Bode might be a promising way to unify MERGE and labelling theory. For instance, Bode (2020:108) argues that '[n]ot labeling all copies at once, would "turn" the instances of "John" [...] into repetitions, which, in

⁸⁹ What this would mean is also unclear: would it involve all the features, including phonological and semantic ones?

standard terms, yields a violation of the Theta-criterion at SEM'. This is very similar to Chomsky's discussion of theta-theory constraining the output of FC.

Bode's (2020) primary concern is the treatment of adjuncts in minimalism. Her approach centres around equating labelling with Transfer. She argues this using the following logic:

- i. Transfer makes objects inaccessible to syntactic operations.
- ii. Labels are not syntactic objects. Hence labels are inaccessible for syntax.
- iii. Labelling makes objects inaccessible to syntactic operations.
- iv. **Labelling = Transfer** (Bode, 2020:97)

iii. is not standard. The idea is that labels are inaccessible to the syntax, which means that labelled elements themselves become inaccessible. However, this is not necessarily an automatic consequence, which I discuss below.

Labelling/Transfer makes syntactic objects (SOs) invisible to the syntax and visible to the interfaces. Given that labelling and Transfer are equated, Bode proposes a unique solution to labelling conflicts, which are resolved under i. or ii.:

- i. Label XP or YP immediately after External Merge.
- ii. Label XP or YP immediately after Internal Merge. (Bode, 2020:111)

Adjuncts, which create labelling conflicts, are labelled/transferred upon Merge (IM or EM) and so are automatically invisible to the syntax.

Clarification is required concerning the mechanisms surrounding IM and copies.

Bode argues that, when an element moves, both it and its lower copy are labelled/transferred simultaneously. Indeed, as mentioned in section 6.4.1, copies are the same element, so it is undesirable for them to behave differently. Thus, when

an SO moves, both the higher and lower copy become invisible (section 6.5.3 discusses successive-cyclicity):

the main idea of copies is that they are a single, discontinuous object. We therefore further argue that all copies get labeled simultaneously rendering them invisible to syntax and distinguishing them from repetitions at the interfaces. That means that *identical elements which are labelled (= transferred) at the same time are copies [...] and not repetitions* (Bode, 2020:101, emphasis mine)

With both copies rendered invisible, there is no need for feature sharing, the structure is labelled in the traditional way.

Interestingly, Bode's approach requires a copy formation operation to identify copies as being the same element. This is not mentioned in Bode and is a problem with her approach, yet, if we assume FC-theory, the two approaches integrate well. Bode, working outside of FC-theory, only considers IM-copies, but this approach can also apply to EM-copies.⁹⁰

For Bode, Transfer and labelling are equated. I continue to assume that these are two separate notions, because, aside from the fact that equating the two is not standard, it appears that labelling and Transfer play different roles. Indeed, the fact that labelling is related to the interfaces does not necessarily imply that it transfers structure to the interfaces. Since it is possible to conceive of transferring unlabelled structures,⁹¹ the two phenomena appear to be separate. Furthermore, Bode's approach seems to imply that the clausal spine can never be labelled. For instance,

⁹⁰ This requires a slight rephrasing of Bode's labelling conflict resolution; however, this is necessary regardless given that derivations are Markovian.

⁹¹ Bode (2020:125) discusses such cases, deriving island effects from this paradox.

in (116), the adjunct is transferred upon labelling. However, it is unclear how the entire structure can be labelled, given that {read a book} is also transferred upon labelling.

(116) {{read a book} {in which garden}}

Instead, I adapt from Bode the notion of transfer of copies. This would alter (109) to (117).

(117) FC transfers copies.

As for Bode, copies are no longer rendered invisible. Rather, they have been transferred before the rest of the structure, so they are not available at the time of labelling.

This approach is not without its problems. First, (117) is a stipulation. FC is a copy formation operation; it is unclear why it would be involved in Transfer. Second, this approach involves stating that FC transfers copies, but otherwise, Transfer happens at the phase level. This is problematic, as C-I would have to reconstruct the structure in a rather complex way. This approach to structure is certainly not ‘an optimal solution to legibility conditions’ (Chomsky, 2000:96).⁹²

6.5.2 Alternative solutions

Given this issue concerning Transfer, this section sketches two alternative solutions. The first one is presented in (118).

(118) FC endows both copies with a ‘copy’ feature, which states that they are irrelevant for labelling.

⁹² Bode’s approach faces a similar issue, though the details differ given that she equates labelling and Transfer.

FC must in some way modify the elements it relates to make it clear that they are copies. The exact way in which this is done is left open: could it be via a feature? This would constitute a worse violation of the NTC than the usual application of FC, as it involves features, which are syntax-internal. Furthermore, the notion of a ‘copy’ feature, which states that the constituents are irrelevant for labelling, is undesirable. Indeed, nothing about the copy relation naturally entails invisibility, and ‘copy’ is not a feature we would want to allow.

Additionally, this approach suffers from the same semantic issue as labelling theory. If copies are given a feature which states that they are invisible for labelling, it is unclear how they can then be visible at C-I.

(118), then, is fundamentally flawed. (119) may be somewhat less problematic, although still not ideal.

(119) FC weakens the labels of copies.

(119) relies on the notion from labelling theory that some elements, for instance conjunctions, are too weak to label (Chomsky, 2013:46). FC would somehow weaken the labels of the copies, rendering them too weak to label. However, this suffers from the same issue as Chomsky (2013, 2015): it is unclear how labels can be too weak to produce labelling conflicts, but strong enough to provide an interpretation at C-I. Additionally, it is unclear why FC would weaken labels, this would have to be stipulated. The strengthening of labels has been suggested, but label weakening is new. Lastly, the precise notion of ‘weak’ and ‘strong’ is never defined clearly in labelling theory or other work referring to feature strength.

One further issue must be discussed in relation to these three approaches, since they reject the idea of feature sharing. Indeed, if feature sharing is no longer a

possibility, {XP, YP} structures which do not involve an FC-relation cannot be labelled. For instance, the indirect object DP and VP in (120) form a symmetric structure which cannot be resolved through FC. This question requires further research.

(120) {_{TP} {_{DP} he} {_{TP} T {_{VP} {_{DP} he} {_{VP} gives {_α {_{DP} her} {_{VP} gives {_{DP} a gift}}}}}}}}

6.5.3 Successive-cyclicity

The above sections have illustrated some possible solutions to lower copy invisibility and feature sharing, with my approaches presenting significant and similar issues, and my adaptation of Bode's (2020) approach struggling, given Transfer.

Successive-cyclicity poses problems for all three approaches: any cases of movement, or EM-FC relations involving more than one step, are an issue. If both copies become invisible after FC, it is unclear how the next copy link is to be formed.

Bode's (2020) solution is to state that copies are only labelled/transferred when the highest copy is reached. However, this requires look-ahead to locate the highest copy, which is undesirable. Furthermore, my adaptation of Bode adopts phase theory, so there would be a risk of inscriptions being transferred before having been assigned a copy relation.

Two alternative solutions seem possible. Firstly, FC could transfer copies, as in (117), but leave behind a 'trace' with which the next higher copy can form an FC-relation. This, however, not only violates the NTC, but is extremely unhelpful for the semantics. It is essentially an undesirable return to trace theory.

Alternatively, there could be a delay in Transfer (/weakening/assignment of the 'invisible copy' feature) for the higher copy. FC always transfers (/weakens/assigns a feature to) the lower copy, but the higher one is not transferred (/weakened/assigned

the feature) until the next FC-relation, where it becomes the lower copy. This is similar to the notion in phase theory of the ‘higher’ elements, i.e., those at the edge, not being transferred until the next phase. To avoid feature sharing, a rule must be stipulated for the highest copy, stating that it is also transferred; this is analogous to the stipulation that the edge of CP gets transferred once the sentence is completed. This approach is not compatible with the idea of copies, being the same element, behaving in the same way (see section 6.5.1).

6.5.4 Interim summary

Labelling theory can be unified with MERGE under the assumption that FC renders copies invisible for labelling. However, labels are claimed to pertain to interpretation, yet, in practice, primarily serve the syntax. This is obvious when considering semantic issues relating to lower copy invisibility and feature sharing. I have attempted to provide tentative solutions to these problems, but this has proven unsuccessful; I argue that this stems from the inherent contradiction within labelling theory about the status of labels in relation to the syntax.

Thus, although FC-theory and labelling theory can be unified, the fundamental problems with both remain. The final section illustrates these issues and other challenges arising when unifying FC-theory and labelling, by examining structures previously analysed using FC and adapting them to my framework.

6.6 Illustrations

This section illustrates the derivation of various structures under the assumption that FC renders all copies invisible for labelling, for instance under Transfer as in (117).⁹³

This has similar predictions to standard labelling theory, with crucial differences being the absence of feature sharing and the existence of labelling effects in M-gap structures. I argue that FC-theory and labelling can be unified, but that this does not detract from the fundamental issues facing both theories.

I do not consider the mechanics of head movement in these structures. Head movement, if considered a syntactic phenomenon, seems to pose a problem for labelling theory. Indeed, if copies become invisible after movement/FC, it is unclear how heads which move can ever label the structure. For instance, in (121), it is unclear how *v* can label *vP* if it has moved, given lower copy invisibility.

(121) {_{TP} T {_{vP} *v* ...}}

This issue appears related to the question of when exactly labelling occurs.

Chomsky argues that labelling takes place at the phase level; however, it is not obvious whether this refers to labelling of the phase head complement or the entire phase (including the head and edge). The correct labelling is derivable for head movement only if the entire phase is labelled here. In Chomsky's work, it appears that the edge is not labelled at this point⁹⁴ (the edge of *vP* can only be labelled after DP movement in a *vISH* structure), suggesting that this is an issue.⁹⁵ Given that Chomsky relegates head movement to externalisation, this issue does not appear to

⁹³ The illustrations of the structures are identical regardless of which of the three approaches are used; I assume (117) given that it is grounded in prior research.

⁹⁴ The labelling of elements which are not transferred until a higher phase would commit to labelling being syntactic.

⁹⁵ T-to-C movement is problematic regardless of when labelling occurs given that T is not a phase.

have been discussed in depth; work which incorporates head movement and labelling seems to involve the assumption that labelling happens immediately, rather than at the phase level (e.g., Moro & Roberts, 2023:8, where a head moves, but the lower structure is still labelled by this head).⁹⁶ Further discussion is beyond the scope of this work,⁹⁷ and head movement in the following examples is not addressed.

6.6.1 Control

The most obvious example of a derivation which has been analysed using FC is Control. As mentioned in section 6.3, the merger of the controllee creates a symmetric structure if it is analysed as a phrase. In labelling theory, Chomsky must, therefore, analyse PRO as a head; feature sharing is not possible here given the absence of Agree. In FC-theory, by contrast, the controllee is necessarily a phrase.

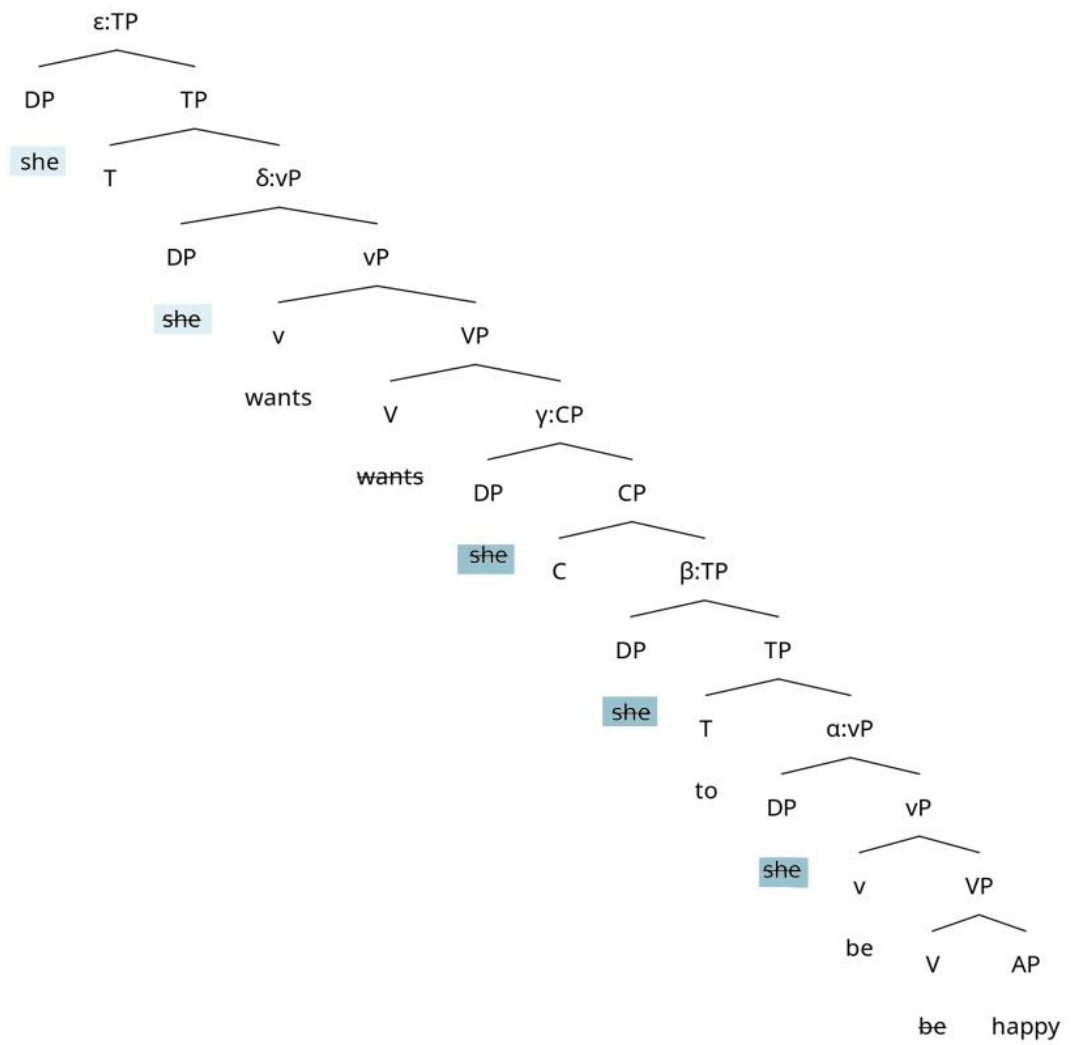
Subject Control and Raising to Subject derivations have analogous structures in my approach, as indicated in (122) and (123) (I provide a detailed derivation of (122) below). The symmetric structures are resolved by the application of FC, which transfers the copies, and the structure is labelled by the remaining elements. FC'ed elements are highlighted. The different colours indicate the two elements separately EM'ed into the structure (in accordance with Duality₂) and their IM-copies: this distinction is now irrelevant in the syntax but is indicated for readability. Object Control and Raising to Object (ECM) derivations are also analysed with analogous structures.

⁹⁶ See also Bode (2020:142-143) for an alternative approach to head movement in the context of labelling = Transfer.

⁹⁷ Dadan (2019:82) discusses a different problem with labelling in head movement structures: application of head movement results in a symmetric {X, Y} structure. This issue is absent in my approach given that I assume that both copies are invisible for labelling, yet remains an issue for standard labelling theory.

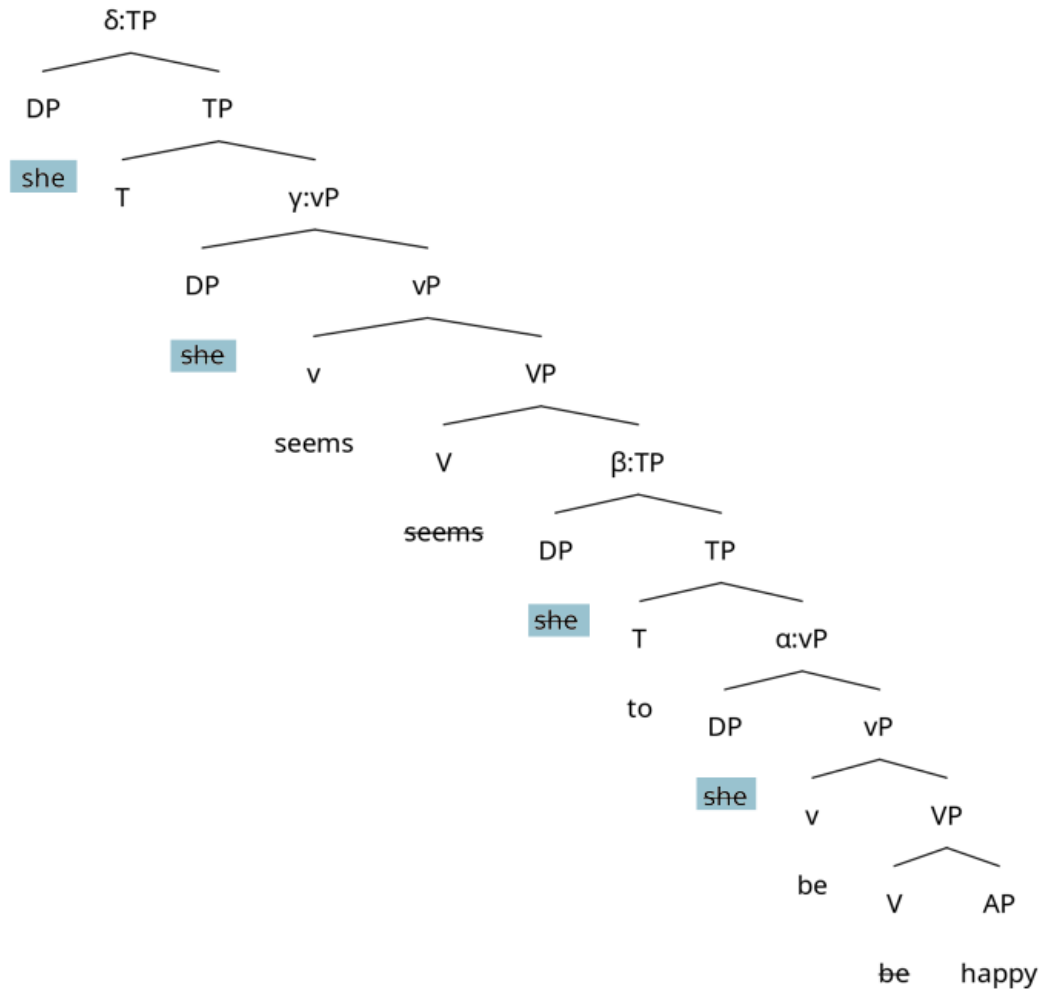
(122) She wants to be happy

Subject Control



(123) She seems to be happy

Raising to Subject



For clarity, I illustrate (122). First, vP is created, and *she* is merged with vP, forming α. Next, *to* is merged, and *she* is IM'ed, forming β. Subsequently, FC applies, transferring the lower inscription of *she*. α is labelled vP. C is merged, and *she* is IM'ed, forming γ. β is labelled TP, as the lower *she* is transferred upon FC. After the merger of *wants* and v, *she* is EM'ed, forming [δ DP vP]. Thus, γ is labelled CP, with FC transferring the lower inscription of *she*. T is merged, and *she* is IM'ed, forming [ε DP TP]. FC applies, and δ is labelled vP. ε is labelled TP by the stipulation that the highest copy must be transferred, akin to the process which transfers the edge of CP once a sentence is completed (see section 6.5.3).

Control derivations are analysed with a CP layer (see section 4.4.2). Therefore, *she* must be IM'ed in Spec CP, or FC cannot apply as the relevant inscriptions are separated by a phase boundary. This holds both with the PIC₁ (the version used here and in MERGE) and the PIC₂ (see footnote 53). This movement instantiates improper movement. As discussed in section 4.4.2, this is a challenge for FC-theory, which requires further research.

6.6.2 Coordination

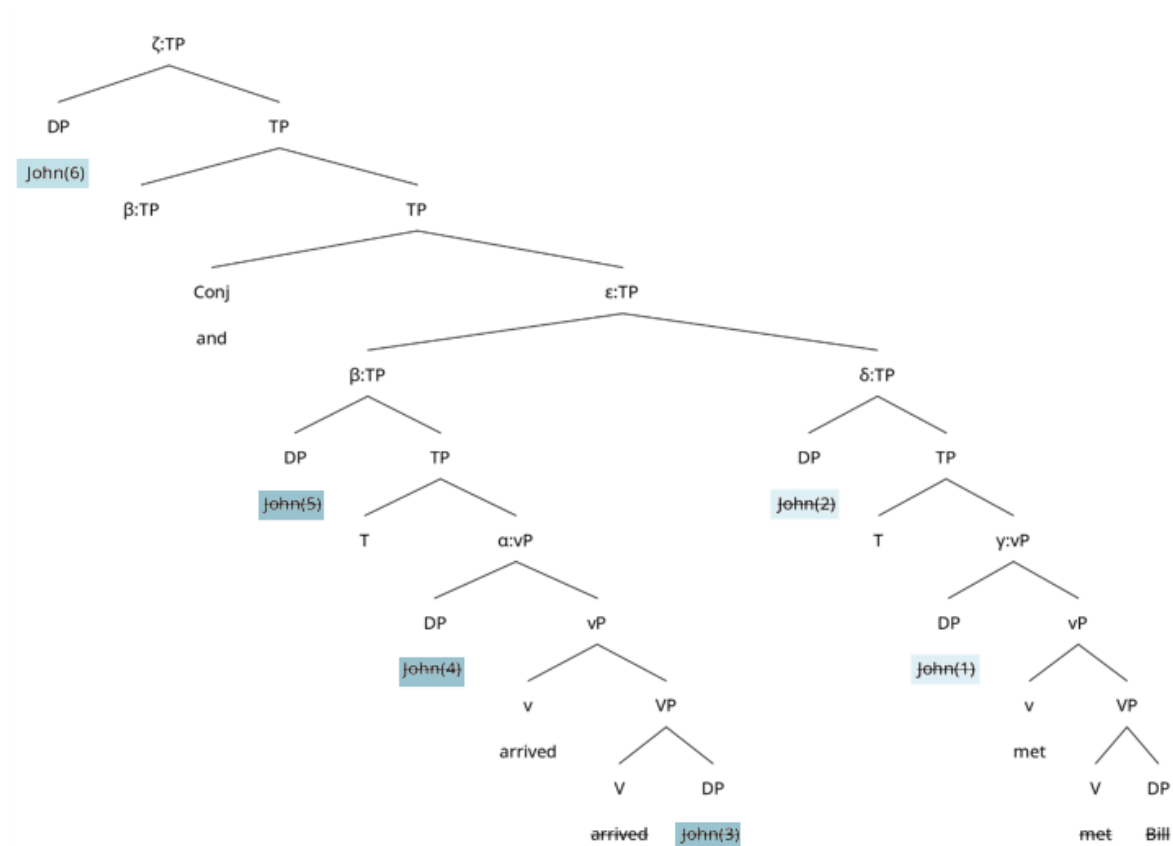
Kitahara (2023) uses coordination to illustrate the application of FC. He argues that coordination structures are derived as in (124), with *John*₁ and *John*₂ EM'ed in theta-positions according to Duality₂, and *John*₃ IM'ed to a non-theta-position.⁹⁸ Kitahara argues that FC must apply for the higher copy to be theta-linked to a theta-position. Section 5.1.2 argued that FC does not involve theta-linking, but there would still be a theta-theory violation here in the absence of FC.

(124) *John*₃ {{arrive *John*₂} and {*John*₁ {meet Bill}}}} (Kitahara, 2023:217)

(125) applies my labelling approach to this structure, assuming (124) involves TP-coordination.

⁹⁸ For Kitahara, this is forced by efficient computation rather than Duality as he uses definition (33) of Duality₂; see section 3.4.

(125) John arrived and met Bill



*John*₁ and *John*₃ are EM'ed in theta-positions and move to what will become Spec-TP, creating four symmetric structures. This is followed by EM of β with δ, forming ε, which is merged with *and*. The unlabelled structure β is then IM'ed. Lastly, *John* moves to the top of the structure. Crucially, Kitahara argues that either *John*₂ or *John*₅ can move. FC then applies, forming a copy relation between all inscriptions of *John* regardless of whether movement was involved. This in turn resolves the labelling conflicts α-δ and ζ. FC applies between both copies of β (now TP), rendering both copies inaccessible for labelling, so ε is labelled TP by δ=TP.

Many concerns arise. Firstly, Kitahara's analysis of coordination seems to be a preliminary outline rather than a fully developed analysis. A discussion of whether this is a satisfactory account of coordination is beyond the scope of this work.

However, it is important to note that the only FC-based approach to coordination currently available is problematic.

Secondly, this derivation does not appear to derive the correct linear structure: *and* c-commands *arrive* even after movement of $\beta=TP$, yielding **John and arrived met Bill*.

Finally, (126) appears pragmatically marked, whereas (127) is completely ungrammatical, which cannot be accounted for in this approach.

(126) John₁ arrived and John₁ met Bill

(127) *John arrived John and met Bill

Additionally, various issues arise when adapting Kitahara's approach to labelling. Firstly, it is unclear how to label the higher half of the structure, since conjunctions are, for Chomsky, too weak to label. Indeed, labelling the structure ConjP seems unhelpful for the semantics. If Conj cannot label, and MS continues, the structure is presumably labelled TP by T in $\delta=TP$.

Secondly, movement of β is required to yield the correct derivation. Movement of unlabelled structures is controversial (Chomsky, 1995a).

Thus, Kitahara's (2023) analysis of coordination faces significant issues, both on its own and when adapted to labelling theory. This is understandable since it appears to be a sketch. Further research is required to develop a satisfactory theory of coordination in FC- and labelling theory.

6.6.3 Saito (2022)

This section discusses Saito's (2002) analysis of (128). Saito argues that FC can derive configurations which would otherwise be analysed with sideways movement.

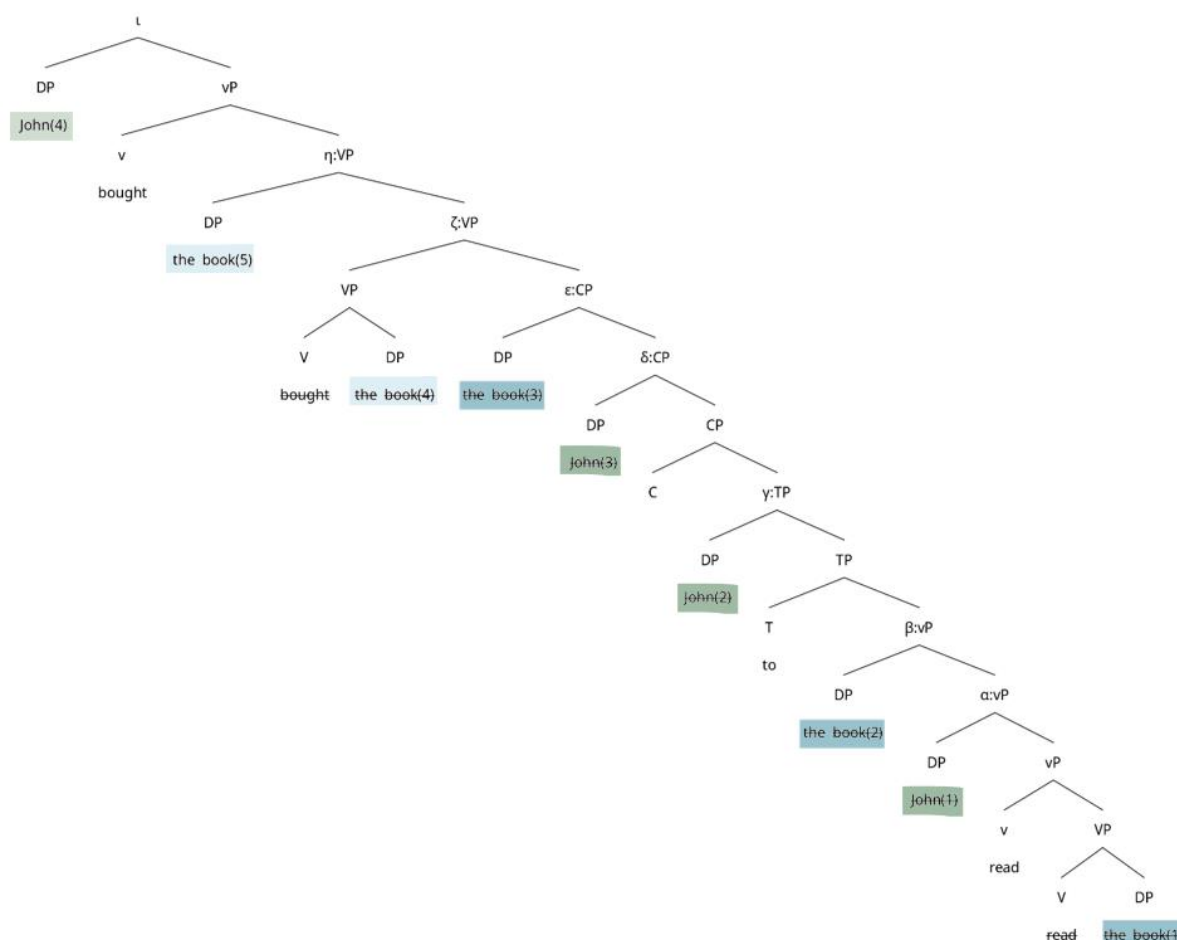
Saito applies FC to various structures, including depictive secondary predicates, resultative small clauses, parasitic gaps, and structures typically analysed with null operators. I focus on the latter.

Saito (2022:170) adapts Hornstein's (2001) analysis to derive (128) without using a silent operator, as is standard. (129) presents my adaptation. FC resolves most of the symmetric structures straightforwardly.⁹⁹

⁹⁹ ζ is a problem, however, which relates to the status of head movement concerning labelling. ι represents the labelling conflict which will be resolved after vISH.

(128) John bought the book to read

(129)



FC applies between *the book₃* (EM'ed in a theta-position) and *the book₄* (IM'ed in a non-theta-position), deriving a sideways movement-like configuration. Allowing FC to yield sideways movement-like derivations seems problematic, as it may allow island violations, for instance. Furthermore, FC requires c-command, and *the book₃* and *the book₄* do not c-command each other. See section 5.2.1 for related discussion.

Lastly, this structure presents problems concerning intervention. Both *John* and *the book* should be interveners for each other in many parts of this structure, including movement through both phase edges. For instance, both *John₂* and *the book₂* must

move to Spec-CP for FC to apply.¹⁰⁰ The question of intervention effects may also be relevant for the FC relations in the higher part of the clause.

Thus, although Saito's approach to (128) can be adapted to labelling theory without significant issues, the analysis remains problematic in other regards, in particular concerning FC-relations.

6.6.4 Wh-dependencies

Chapter 5 discussed Manzini & Rugna's (2024) application of FC-theory to wh-dependencies in Germanic. This section returns to these in the context of labelling. Manzini and Rugna's work uses standard labelling theory, despite the issues which I have argued arise when attempting to integrate FC-theory with labelling theory.

Manzini and Rugna argue that EM is forced in wh-doubling structures, as the lower inscriptions participate in labelling by feature sharing, rendering them essentially frozen. This approach entails that structures receive different labels depending on whether the higher copy is EM'ed or IM'ed, which seems undesirable, given that the difference between these structures is syntactic rather than semantic. For instance, the lower clause in a wh-movement structure (130) receives the label CP (via lower copy invisibility), whereas the corresponding clause in a wh-copying structure (131) is labelled <F,F> (via feature sharing).

(130) Wen glaubst du dass Irina liebt?

who believe you that Irina loves

'Who do you believe that Irina loves?'

{<F,F> wen glaubst {IP du {VP wen glaubst {CP wen dass Irina liebt wen}}}}

¹⁰⁰ Movement of *the book*₄ is also necessary to derive the correct configuration for FC.

(131) Wen glaubst du wen Irina liebt? (=88))

who believe you who Irina loves

‘Who do you believe that Irina loves?’

{<F,F> wen glaubst {IP du {VP wen glaubst {<F,F> wen Irina liebt wen}}}}

(Manzini & Rugna, 2024:11-12)

Other than the semantic issues surrounding this, the approach requires

Agree/labelling via feature sharing to be optional (Manzini & Rugna, 2024:12),

limiting the predictive power of the analysis. The inconsistencies between FC-theory

and labelling theory are also highlighted: labelling cannot distinguish IM- and EM-

copies, so they should behave identically regarding labelling.¹⁰¹

In my approach, these problems are avoided. The symmetric structure is resolved in

both cases by application of FC, yielding the traditional labelling.

(132) Wen glaubst du wen Irina liebt?

who believe you who Irina loves

‘Who do you believe that Irina loves?’

{CP wen glaubst {TP du {VP wen glaubst {CP wen Irina liebt wen}}}}

Thus, it is possible to adapt Manzini and Rugna’s analysis for improved consistency

with FC-theory, while mitigating the labelling problems. Nevertheless, other issues

concerning Manzini and Rugna’s analysis remain, particularly copy visibility and

identity (see chapter 5). Relatedly, the idea of FC rendering lower copies invisible for

¹⁰¹ There is one advantage in labelling the structure <F,F>, namely that it ensures that the embedded clause is not labelled as an interrogative. Indeed, IM of a wh-phrase into Spec CP is considered a problem, as *glauben* (‘believe’) selects a declarative complement. <F,F>, however, cannot involve wh-feature sharing, as *glauben* does not have such a feature. Instead, Manzini & Rugna (2024:13) argue that <F,F> represents Force features, the interpretation of which can be determined by context, and so are not interrogative in the embedded clause. Nevertheless, this approach does not explain wh-movement structures, and <F,F> is still not a particularly useful label.

labelling is particularly problematic for Rugna and Manzini's structures. Indeed, EM'ed FC-copies can be non-identical for Rugna and Manzini (see section 5.3.1). Rendering non-identical lower copies invisible is particularly problematic. This issue, for which I have provided unsatisfactory tentative solutions (e.g., by suggesting that FC transfers copies, rather than actually rendering them invisible) is particularly stark here.

6.7 Summary

Thus, I have argued that labelling theory can be unified with MERGE, with similar empirical predictions. I have adapted some structures which have been analysed using FC to my FC-based labelling approach.¹⁰² However, the fact that this is possible does not detract from the fundamental issues facing both theories. This has become evident in the various outstanding issues in the structures have discussed.

Labelling theory faces several significant challenges, and these persist in my FC-based approach. In particular, feature sharing and lower copy invisibility present significant semantic problems, which I argue illustrate a fundamental issue with labelling theory. Indeed, while labels are claimed to be necessary for the interfaces, they are in practice primarily used to yield the correct syntactic derivations. Potential solutions, for instance stipulating that FC transfers copies, face significant fundamental challenges; ultimately, the notion of copy invisibility in any form remains a stipulation existing solely to produce the correct derivations.

¹⁰² FC has been applied to Latin (Jakielaszek, 2023), long-distance relative clauses in Japanese (Hayashi, 2022), Japanese serial verb constructions (Fong et al., 2023) and even Agree (Roberts, 2024), which I cannot discuss for space reasons.

7 Conclusion

Minimalist syntax aims to maximise theoretical elegance and explanatory adequacy. MERGE is a development in this direction, but, given its novel nature, many issues remain. This thesis has examined various problems which arise when scrutinising MERGE, including its interaction with pre-existing elements in the theory. I have focussed on the duality of semantics, FormCopy, and the interactions between MERGE and labelling theory, arguing that labelling theory must be rethought. Indeed, given that copies no longer arise automatically from IM, it must be FC, not movement, which makes them invisible for labelling. However, even if labelling theory is altered in this way, fundamental issues remain with both labelling theory and FC, for which no ready solutions seem to be apparent.

Chapters 2 and 3 discussed and scrutinised the various definitions of the duality of semantics proposed thus far. Duality₁ struggles to account for the EM of discourse-related elements. Duality₂ encounters issues with the EM of non-arguments. Furthermore, both versions of Duality have been associated with the A/A' distinction, which I have argued is problematic. Lastly, it is unclear how Duality₂ can be used to constrain MERGE, given that theta-roles are assigned configurationally and are not present in the syntax. This recurring pattern of trying to simplify the syntax by shifting as many phenomena as possible onto the interfaces results in inconsistencies in the extent to which these phenomena are considered to belong in the syntax.

Chapter 4 discussed Chomsky's (2019b) use of Duality₂ to derive the copy/repetition distinction. Chomsky's (2021) work introduces FC, which is intended to derive the Control/Raising distinction. Empirical issues with this approach to Control were

discussed, with a similar issue arising as for Duality₂: the Control/Raising distinction appears to be syntactic, yet can only be determined at interpretation.

The introduction of FC impacts the understanding of ‘copies’ in the theory, including rendering the copy/repetition distinction from Chomsky (2019b) obsolete. Chapter 5 discussed some theoretical problems facing FC, particularly given the introduction of Preservation, which keeps track of IM-copy relations. This entails that there are two elements in the theory which refer to ‘copies’. The existence of both Preservation and FC appears redundant, and introduces complications to the theory, as well as confusion when interpreting Chomsky’s work. I argue that copy theory in MERGE involves at least two distinct types of copies, namely FC-copies and Preservation-copies. I illustrate the proliferation of different types of ‘copies’ now present in the theory, which presents a highly undesirable complication. I claim that the type of copy referred to in various situations must be clarified, as the consequences differ. I illustrated this with the MS approach to IM not violating MY, arguing that this no longer follows automatically. Lastly, chapter 5 discussed the Markovian nature of derivations, arguing that it is essential to clarify which phenomena are syntactic as opposed to interpretive, as this affects whether they are Markovian and thus which elements they can access. This underlines the problem of the ambiguous status of various phenomena discussed in the previous two sections.

Labelling has a particularly unclear status in relation to the syntax, as discussed in chapter 6. I argue that labelling is a syntactic phenomenon in Chomsky’s work, despite his claims to the contrary. It is, therefore, Markovian, and can only refer to FC-copies. Thus, the claim in labelling theory that lower copies generated by movement are invisible no longer follows. Instead, I suggest that FC renders copies invisible for labelling, resolving labelling conflicts. In this way, FC-theory and labelling

theory can be unified, though this fails to resolve or even address any of the fundamental issues with labelling theory. I have attempted, unsuccessfully, to bypass the semantic problems concerning lower copy invisibility by providing tentative alternatives. Ultimately, lower copy invisibility remains a stipulation existing only to yield the correct derivations. I argue that this highlights the ambiguous status of labelling as a phenomenon claimed to be necessary for the semantics, yet in practice only serving the syntax.

Many questions remain. FC-theory requires further study, both from empirical and theoretical perspectives. It remains to be seen whether the numerous issues with this new theory may be resolved or if it is inherently flawed. One outstanding question concerns the status of binding theory. It is unclear how Binding Principle C can be reconciled with an FC-based approach, which ties in to the question of how FC might be used to derive coindexation, as in Manzini & Rugna's (2024) work.

Additionally, given the significant concerns and complexity of copy theory in MERGE, it may be worth considering whether a return to trace theory could offer a more viable alternative. In section 6.5.3, I suggested that a framework where FC transfers copies, leaving behind a trace in the process, would constitute an undesirable return to trace theory. However, it could be worth exploring whether such an approach might prove more effective than copy theory in MERGE. For instance, one might consider a scenario where, when an element moves, its original inscription is transferred, leaving behind a trace. In other words, movement must be followed by Transfer of the lower copy.¹⁰³ The information contained in the lower copy would not be lost, it would simply be inaccessible to the syntax. This could also explain lower

¹⁰³ Note that such an idea has some similarities with those discussed in section 5.6.2.

copy invisibility for labelling: it does not seem implausible to assume that traces cannot participate in labelling. This might be an avenue for further research.

Section 1.1.6 discussed which operations are allowed in MERGE. Pair-Merge, still included in Chomsky's theory, would ideally be eliminated, although how remains unclear. For relevant ideas, see Bode (2020). The status of head movement requires further research, as highlighted in various places throughout this work. Roberts (2024) suggests an approach linking FC and head movement. Lastly, Chomsky is unclear on the status of Agree in MERGE. It remains necessary for the syntax given the existence of feature sharing, but Chomsky suggests relegating it to externalisation. Roberts (2024) is also relevant here, as he suggests that FC might derive Agree.¹⁰⁴

Another question concerns the nature of MERGE: how restrictive should it be, and what constraints should be imposed on it? Section 1.1 discusses how MERGE itself is (133), but that it is constrained by third-factor and interface conditions to yield the desired outcome.

$$(133) \text{ Merge}(P_1, \dots, P_m, WS) = WS' = [\{P_1, \dots, P_m\}, \dots] (= (2))$$

Thus, MERGE is not entirely free, and some derivations are expected to be ruled out by Merge, including extensions of Merge, which are ruled out by MY, and violations of Duality₂, since this is argued to constrain MERGE. Chomsky does not wish for the syntax to be crash-proof, yet some phenomena are ruled out by MERGE; determination of the boundary requires further consideration.¹⁰⁵ Furthermore,

¹⁰⁴ Manzini & Rugna's (2024) approach to FC involves feature sharing. Feature sharing is closely tied to Agree, as seen. Thus, it appears that two papers have in some sense suggested that FC could be linked to Agree, which indicates that this might be an interesting direction for further research.

¹⁰⁵ One could argue that MERGE itself as in (133) is entirely free, i.e., the elements which constrain MERGE, such as MY and Duality, are not actually a part of MERGE.

MERGE is not entirely free in the sense of applying without trigger: I have argued that labelling still acts as a trigger for MERGE/FC.

Lastly, issues arise when aiming to maximally simplify MERGE, as this places a substantial burden on the third factor and interfaces. I have already discussed how this is an issue for the interfaces in terms of the ambiguous status of various phenomena. Furthermore, this theory implies that most structures generated by the syntax will be rejected at C-I, which is hardly 'an optimal solution to legibility conditions' (Chomsky, 2000:96). Moreover, syntactic selection cannot be reduced to S-selection (Grimshaw, 1979), and if ungrammaticality is determined at C-I, it is unclear how to distinguish ungrammatical sentences from those which are semantically nonsensical. Concerning third-factor elements, theories such as MERGE risk positing third-factor operations without proper justification. It is essential to properly justify how an operation falls under the third factor for the argument to be valid. FC poses a problem here, as it is unclear how it is a general law of nature or a principle relating to efficient computation. Further research is required to render MERGE theoretically simple without compromising on empirical coverage, acquirability, and evolvability.

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